

Original articles

Time-varying volatility model equipped with regime switching factor: Valuation of option price written on energy futures

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ARTICLE INFO

Keywords:

Energy markets

EM algorithm

Option pricing

Regime switching model

ABSTRACT

This paper explores the calculation of European option prices on energy futures using a time-varying volatility model enhanced by a regime switching factor. We develop a semi-analytical method to determine the price of European options on these energy futures, involving the derivation of the characteristic function for the energy futures' dynamics. To determine the parameters of the regime switching model and identify when economic states change, we employ the EM algorithm, utilizing real gas futures price data. We validate our closed-form solution for the option pricing through simulations employing the generalized antithetic variates Monte-Carlo technique. A comprehensive numerical analysis demonstrates the effectiveness of our proposed methodology.

1. Introduction

Regime switching models have gained widespread popularity and practical utility in the fields of econometrics and finance, with their origins traceable to the pioneering work of [1]. These models were further popularized in economic and financial applications by [2,3]. A defining characteristic of these models is their capacity to adapt model dynamics over time in alignment with the states of an underlying hidden Markov chain, also referred to as a modulating Markov chain. This feature offers a natural and efficient way to model how pricing series are impacted by structural economic shifts, such as economic expansions or financial crises. Over the last two decades, the valuation of contingent claims within regime switching frameworks has gained significant attention in both finance and actuarial science. The primary appeal of applying regime switching models to option valuation lies in their ability to incorporate condition shifts, such as changes in economic conditions, that impact asset price dynamics (see [4]). Furthermore, these models have been shown to capture important empirical characteristics observed in options pricing data, such as the implied volatility smile or smirk (see [5,6]). Recent advancements in option pricing under regime switching models include contributions from several researchers, among others [7–10], and [11].

In recent years, the valuation of energy futures has gained significant attention. A distinctive feature of energy prices is the recurrent presence of jumps over time, which, however, tend to revert to the mean level swiftly. Regime switching models, capable of accommodating these events through different economic states, play a crucial role in representing energy price dynamics (see [12]). The literature offers a variety of regime switching models tailored for analyzing energy dynamics to assess futures values. Recent examples include [13,14], and [15]. Unlike stock markets, where options are commonly based on spot prices, energy markets, particularly for electricity and gas, feature options traded on energy futures. European-style options written on energy futures contracts are common in these markets. However, trading of Asian-type options at the Nord Pool electricity exchange existed during

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the 1990s but was discontinued in 2000. Recent articles on the pricing of options on energy and commodity futures, within the regime switching model, include works by [16–22], among others.

The traditional assumption of constant volatility and correlations significantly contrasts with empirical evidence. Recognizing this disparity, recent models have shifted towards adopting time-varying volatility, incorporating elements like conditional heteroskedasticity, as seen in [23] with dynamic copulas, and in [24] with co-integrated asset prices. However, a common limitation among these studies is their approach to risk neutrality, which is typically conducted on an individual stock basis. Given the well-documented correlation between financial assets, the notion that exposure to this risk is cost-free appears counterintuitive, clashing with core financial theories such as the modern portfolio theory and the capital asset pricing model. Furthermore, while these papers predominantly rely on a Gaussian model, empirical data often reveal that financial asset returns exhibit skewness and leptokurtosis, challenging the validity of such an oversimplistic assumption (see [25]). Historical models with constant volatility suggest that logarithmic asset price returns follow a Gaussian distribution. Yet, empirical data suggest that logarithmic stock returns deviate towards a non-Gaussian distribution (see [26]). Incorporating a regime switching mechanism offers a pathway towards achieving a non-Gaussian model, as discussed in [6,27], and [28]. In this study, we aim to bridge the gap between theoretical modeling and market reality by integrating a regime switching factor into the time-varying volatility model. This represents a novel attempt to more accurately reflect future prices in line with market observations. This paper addresses the challenging process of calibration within the regime switching model framework. Due to the model's latent switching mechanism, where parameters shift in response to abrupt temporal changes, it is imperative to simultaneously estimate these parameters along with the state values of the hidden Markov chain process (see [29]). We employ the method proposed by [30], utilizing the expectation–maximization (EM) algorithm to estimate the parameters of the hidden Markov model. Empirical data support our regime switching model, accurately capturing the dynamics of energy futures prices through our calibration.

Let F_t denotes the futures energy price at time t with maturity date T . The dynamics of the futures price process $t \mapsto F_t$, for $t \leq T$, have been modeled in [31] by the stochastic differential equation

$$dF_t = F_t e^{-\kappa(T-t)} \xi dB_t, \quad \text{with } F_0 = f > 0, \quad (1)$$

where ξ is the spot volatility, κ is the mean reversion speed, and B_t is a standard Brownian motion under the physical probability measure P . This formulation describes a mean-reverting process where the futures price behaves similarly to a geometric Brownian motion but with a volatility that varies over time. Specifically, the exponential decay term $e^{-\kappa(T-t)}$ implies that the volatility decreases as the maturity T extends further into the future, reflecting the intuition that distant futures prices are more stable. Conversely, as t approaches T , the volatility converges to the spot volatility ξ . Since the returns of the futures price are log-normally distributed at any point before maturity, option prices can be computed analytically. This is achieved by adapting the classical Black model to incorporate a time-dependent volatility

$$\sigma(t, T) = \xi \sqrt{\frac{1 - e^{-2\kappa(T-t)}}{2\kappa(T-t)}}, \quad (2)$$

which replaces the constant volatility parameter of the original formulation. In this mean-reverting framework, the volatility term structure decreases with maturity, and the volatility $\sigma(t, T) \rightarrow 0$ as $T \rightarrow \infty$ indicates that the long-term futures prices tend to exhibit minimal fluctuations. Building on this, the dynamics of the futures price under a time-varying volatility model can be equivalently expressed as

$$dF_t = F_t \sigma(t, T) dB_t, \quad F_0 = f > 0. \quad (3)$$

While the model provided in (3) captures the dynamics of futures prices with time-dependent volatility and mean reversion, extending this structure to a regime switching model can consider various economic states as an exogenous factor. This enhancement enables the model to capture sudden shifts and structural changes in the energy markets that are not addressed by the original formulation. In this study, we implement the regime switching framework for futures price dynamics by allowing the spot volatility parameter ξ on a hidden Markov chain.

Since energy cannot be directly stored, applying a risk-neutral probability measure to energy spot price dynamics is considered meaningless, as highlighted in [32]. In the context of energy markets, the risk-neutral measure is typically defined for the dynamics of futures prices. In an incomplete market described by regime switching models, the presence of an additional risk source tied to regime switches or economic state transitions, as modeled by state transitions of a Markov chain, can complicate matters. As such, the market becomes incomplete. Various techniques exist for selecting a pricing kernel in these incomplete markets. A crucial consideration in regime switching models is whether to choose a pricing kernel that accounts for the regime switching risk. Some studies advocate for pricing this risk (see [33,34]), while others explore scenarios where it is not priced (see [35,36]). One way to address regime switching risk is to integrate it into the pricing kernel or risk-neutral measure. This can be achieved by applying the standard Girsanov transform for Brownian motions and a Girsanov transform for Markov chains to define a product pricing kernel, as demonstrated by [37]. Beyond the risk-neutral condition derived from the equivalent probability measure, additional criteria might be necessary to select a suitable pricing kernel due to the extra source of risk from regime switches or changes in the economic state. However, ideally, a pricing kernel would be determined solely from the risk-neutral condition, eliminating the need for extra criteria. Following an approach analogous to studies such as [38–42], and [43], we use cumulant processes (see [44]) to model energy futures dynamics under a risk-neutral probability measure. These processes, which generalize the cumulant of an infinitely divisible random variable and serve as the exponential compensator of a semimartingale, lead to a generalized Esscher transform.

Over the last decade, the over-the-counter (OTC) market for standard energy indices has seen notable development. Despite this growth, the scarcity of data on OTC option prices presents significant challenges for achieving robust and consistent pricing. Traditionally, one method for valuing energy index options involves replicating the index with a diverse basket of energies, then pricing options on this aggregated basket. Recently, numerous indices have emerged as benchmarks for the overall market, specific sectors, or individual commodities, underscoring the need for a broad approach to harnesses the full investment potential of energy as an asset class. Although several models exist for pricing options on energy futures, research on models tailored for energy indices remains limited, leaving the valuation of such derivatives largely unresolved. To derive a semi-analytical solution for the European option price on energy futures, we employ the Fourier transform. We then validate our formula's accuracy by comparing it with results from the generalized antithetic variates Monte-Carlo simulation method, a recent innovation in variance reduction techniques for pricing Asian options under regime switching models introduced by [43]. This paper aims to bridge this gap by proposing a comprehensive and practical option pricing framework. This framework is designed to address the unique challenges posed by energy futures and the variability of economic states. We want to ensure that the energy futures model aligns with actual market data, and we apply our model to the gas market, comparing our results with those obtained from the classical Black approach. The empirical analysis in this study focuses on gas futures, primarily due to the commodity's inherent volatility, pronounced seasonality, and susceptibility to external shocks such as weather patterns and geopolitical events. These characteristics make natural gas an ideal candidate for regime switching models. However, the proposed framework is not limited to gas markets. It can be readily extended to other commodities, such as electricity and crude oil, which may exhibit distinct volatility regimes driven by factors like supply–demand imbalances, regulatory interventions, and production constraints. Each of these markets presents unique structural features, but the core methodology — modeling regime-dependent dynamics using a Markov-switching process — remains applicable across a broad range of energy and commodity assets.

The paper is structured as follows. In Section 2, we introduce the model dynamics and select a risk-neutral probability measure using the generalized version of the regime switching Esscher transform. Following that, Section 3 utilizes the Fourier transform to derive a semi-analytical option pricing formula. In Section 4, we calibrate the time-varying volatility model, equipped with a regime switching factor, using the EM algorithm and proceed to forecast energy futures using a 1-step ahead method. Section 5 presents the numerical results of the proposed regime switching energy model. The paper concludes with Section 6.

2. The model dynamics

We assume a filtered probability space, $(\Omega, \mathcal{F}, \mathbb{P})$, which is subject to the usual conditions of completeness and right-continuity, over a finite horizon $[0, T]$, where $T > 0$ is the maturity. A standard Brownian motion $B := (B_t)_{t \in [0, T]}$ and an independent hidden continuous-time Markov chain $X := (X_t)_{t \in [0, T]}$ are realized on $(\Omega, \mathcal{F}, \mathbb{P})$. We assume that the Markov chain has two states. From [45], the state space can be represented, without loss of generality, as $\mathcal{E} = \{e_1, e_2\}$, where the vectors e_i form the standard orthonormal basis in $\mathbb{R}^2$, that is e_i is a vector with a 1 in the i th position and 0s elsewhere. In practice, a two-state hidden Markov model is rich enough to capture the empirical phenomena observed in most energy time series. Therefore, our discussion will concentrate on the scenario where the number of states is two. However, our methodology is applicable to an arbitrary number of states.

A business cycle can be divided into two main states, expansionary, and recessionary. When $X_t = e_i$ then $\xi(X_t) = \xi_i$, for $i = 1, 2$. More complex business cycle structures can be captured with a larger state space for X_t . Here

$$X_t = \begin{cases} e_1, & \text{when the energy market is stressed,} \\ e_2, & \text{when the energy market is not stressed.} \end{cases} \quad (4)$$

However, the event $X_t = e_i$ cannot be observed directly, which is why the Markov chain is called hidden. On the other hand, the two-state model is inherently simpler than the H -state model (where $H \geq 3$), as it only allows for direct switches between two regimes. In contrast, the H -state model permits a broader range of transitions among its multiple states.

We denote the rate matrix of X by $\Pi = (\pi_{ij})$ for $i, j = 1, 2$ where π_{ij} is the instantaneous transition intensity of the chain X from state e_i (regime i) to state e_j (regime j). Moreover, for any $i, j = 1, 2$, $\pi_{ij} > 0$ with $i \neq j$, and $\pi_{ii} = -\sum_{j=1, j \neq i}^2 \pi_{ij}$. In this study, we assume that the hidden Markov chain structure has two states, and the rate matrix $\Pi_{2 \times 2}$ contains on the diagonal the negative of the intensities, that is $\pi_{ii} = -\lambda_i$, for $i = 1, 2$, and the off diagonal entries are $\pi_{ij} = \lambda_i, i \neq j$. The transition matrix $\mathcal{P} = (p_{ij})$ contains the probabilities $p_{ij} = \mathbb{P}(X_t = e_j | X_{t-1} = e_i)$ of switching from regime i at time $t-1$ to regime j at time t . Hence,

$$\mathcal{P} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} p_{11} & 1 - p_{11} \\ p_{21} & 1 - p_{21} \end{pmatrix}. \quad (5)$$

Now, let $\bar{\xi} = (\xi_1, \xi_2)^T$, with $\xi_i > 0$ for $i = 1, 2$, represent the market volatility, where $\bar{\xi}^T$ denotes the transpose. It is determined by the regime switching process X and defines the process $\xi := \{\xi_t\}_{t \in [0, T]}$ via

$$\xi_t = \xi(X_t) = \langle \bar{\xi}, X_t \rangle = \sum_{i=1}^2 \xi_i \langle X_t, e_i \rangle. \quad (6)$$

Under the real-world probability measure $\mathbb{P}$, the futures price dynamics are described by the Black model with time-varying volatility and a regime switching factor as follows:

$$dF_t = F_t \sigma(t, T, X_t) dB_t, \quad F_0 = f > 0, \quad (7)$$

where the volatility $\sigma(t, T, X_t)$ is given by

$$\sigma(t, T, X_t) = \xi_t \sqrt{\frac{1 - e^{-2\kappa(T-t)}}{2\kappa(T-t)}}. \quad (8)$$

Thus, if $Y_t = \ln(F_t/F_0)$, then the dynamics of Y_t are

$$dY_t = \sigma(t, T, X_t)dB_t - \frac{1}{2}\sigma^2(t, T, X_t)dt, \quad Y_0 = 0.$$

We now proceed to define the relevant filtrations. Suppose $\{\mathcal{F}_t^X\}_{t \in [0, T]}$ is the right-continuous and $\mathbb{P}$ -complete natural filtration generated by the chain X . Let $\{\mathcal{F}_t^F\}_{t \in [0, T]}$ denotes the right-continuous and $\mathbb{P}$ -complete natural filtration generated by the process F . For each $t \in [0, T]$, we define

$$\mathcal{F}_t := \mathcal{F}_t^F \vee \mathcal{F}_t^X,$$

which denotes the smallest right-continuous, $\mathbb{P}$ -complete filtration generated by $\mathcal{F}_t^F$ and $\mathcal{F}_t^X$.

We assume that each state e_i possesses a distinct volatility level ξ_i . However, it is conceivable that investor activities, characterized by their buying and selling patterns, occasionally result in indistinguishable volatility changes, that is, ξ remains constant (see [46]; [47]). Under such circumstances, discerning a state transition in X_t becomes challenging. It is plausible that alterations in a business cycle's state might be reflected not only in stochastic volatility but also in drift changes. Assuming the volatilities in different states are unique, we can, without loss of generality, infer that X_t is effectively observable. This is because the local quadratic variation of F_t over any small interval preceding t precisely determines $\xi(X_t)$ (see [48]; [49]). As a result, the filtration generated by the process $\{F_t\}$ contains that of $\{X_t\}$.

From [50], the hidden Markov chain X_t satisfies the following semi-martingale representation:

$$dX_t = \Pi X_t dt + dV_t,$$

where $V := \{V_t\}_{t \in [0, T]}$ is a martingale in $(\Omega, \mathcal{F}^X, \mathbb{P})$, and Π is the generator matrix of X . Solving this equation gives

$$X_t = e^{\Pi t} X_0 + \int_0^t e^{\Pi(t-v)} dV_v. \quad (9)$$

Taking expectations under $\mathbb{P}$, and noting that the martingale term has zero expectation, we have

$$\mathbb{E}[X_t] = e^{\Pi t} \mathbb{E}[X_0] = e^{\Pi t} \sum_{i=1}^2 e_i \mathbb{P}(X_0 = e_i),$$

where e_i are the basis vectors corresponding to the states of the Markov chain. Consequently, the expected value of a parameter vector $\tilde{\phi} = (\phi_1, \phi_2)^\top$, which depends on X_t , is

$$\mathbb{E}[\phi(X_t)] = \langle \tilde{\phi}, \mathbb{E}[X_t] \rangle = \langle \tilde{\phi}, e^{\Pi t} \sum_{i=1}^2 e_i \mathbb{P}(X_0 = e_i) \rangle. \quad (10)$$

Throughout this study, the transition time between two states of the Markov chain (or the transition time between two regimes) is assumed to be an exponential random variable with cumulative distribution function $F(x) = 1 - e^{-\lambda(X_t)x}$, where

$$\lambda(X_t) = \langle \bar{\lambda}, X_t \rangle = \sum_{i=1}^2 \lambda_i \langle X_t, e_i \rangle, \quad \lambda_i \geq 0,$$

and $\bar{\lambda} = (\lambda_1, \lambda_2)^\top$ denotes a column vector. Here, $\langle \cdot, \cdot \rangle$ represents the inner product.

Assuming that the transition probabilities from a state $e_i \in \mathcal{E}$ at time t to a state $e_j \in \mathcal{E}$ at time $t+h$ are stationary, we can infer the existence of an infinitesimal generator Π that is independent of time. Let $\mathcal{P}(h)$ be the matrix of transition probabilities defined by $\mathcal{P}_{ij}(h) = \mathbb{P}(X_h = e_j \mid X_0 = e_i)$ for all $e_i, e_j \in \mathcal{E}$. Under this assumption, the transition probability matrix satisfies

$$\frac{d\mathcal{P}(h)}{dh} = \mathcal{P}(h)\Pi = \Pi\mathcal{P}(h),$$

and the entries of $\mathcal{P}(h)$ can be expressed as

$$\mathcal{P}_{ij}(h) = (e^{\Pi h})_{ij}. \quad (11)$$

We analyze the proposed regime switching model by focusing on the Markov process X , which, due to its state-dependent nature, can have transition rate matrices that vary at each jump. This state dependence complicates the use of traditional methods that treat the discrete and continuous components separately, as these approaches are not suitable in this context. To facilitate the analysis, we introduce a state-independent Markov chain $\hat{X} := \{\hat{X}_t\}_{t \geq 0}$ satisfying $X_t \leq \hat{X}_t$ for all time $t \geq 0$. Using this auxiliary process, we establish explicit bounds for the exponential functional $\mathbb{E}\left[e^{\int_0^t \eta(X_s)ds}\right]$, where $\eta(X_t) = \sum_{i=1}^2 \eta_i \langle X_t, e_i \rangle$, and $\eta := (\eta_1, \eta_2)^\top \in \mathbb{R}^2$. To do this, we first provide the following lemma from [51]:

Lemma 2.1. Assume that F_t denotes the commodity futures price at time t , described by model (7). For a positive infinitesimal h , let $\{X_t\}_{t \geq 0}$ be a right-continuous process that represents the two-state hidden Markov chain with values in a finite state space $\mathcal{E}$. The transition probabilities satisfy

$$\mathbb{P}(X_{t+h} = e_j \mid X_t = e_i, F_t = u) = \begin{cases} \pi_{ij}(u)h + o(h), & \text{if } i \neq j, \\ 1 + \pi_{ii}(u)h + o(h), & \text{if } i = j, \end{cases} \quad (12)$$

where the transition rate function is given by $u \mapsto \pi_{ij}(u)$. Then, $\{X_{h\ell}\}_{\ell=0,1,2,\dots}$ is a discrete-time Markov chain with the one-step transition probability matrix $\mathcal{P}_{ij}(h)$ defined in Eq. (11).

For every $u \in \mathbb{R}$, we define a collection of intervals $\{\Pi_{ij}(u)\}_{i,j=1,2}$ on the positive real axis in the following way

$$\begin{aligned} \Pi_{12}(u) &= [0, \pi_{12}(u)], \\ \Pi_{21}(u) &= [\pi_1(u), \pi_1(u) + \pi_{21}(u)], \end{aligned}$$

where $\pi_i(u) = \sum_{j \neq i} \pi_{ij}(u)$. By repeating this construction, we generate a sequence of adjacent intervals $\Pi_{ij}(u)$, each with length $\pi_{ij}(u)$, where the intervals are left-closed and right-open. Next, we introduce a function $\chi : \mathbb{R} \times \{1, 2\} \times \mathbb{R} \rightarrow \mathbb{R}^2$ defined as follows

$$\chi(u, e_i, y) = \sum_{j \in \{1, 2\}} (e_j - e_i) \mathbf{I}_{\Pi_{ij}(u)}(y).$$

Suppose that $H := 2 \max_{i \in \{1, 2\}} \sup_{u \in \mathbb{R}} \pi_i(u) < \infty$, then the process X_t admits the following stochastic differential equation (see [52] or [53]):

$$dX_t = \int_{[0, H] \times [0, H]} \chi(F_t, X_{t-}, y) N(dt, dy). \quad (13)$$

Here, $N(dt, dy)$ represents a Poisson random measure with intensity given by the product measure $dt \times \mu(dy)$, where $\mu(dy)$ is the Lebesgue measure defined on the set $[0, H] \times [0, H]$. Denote by I_t the stationary point process associated with the Poisson random measure $N(dt, dy)$. Since the measure $\mu(dy)$ is finite on $[0, H] \times [0, H]$, the process I_t exhibits only a finite number of jumps within any bounded time interval. Suppose

$$0 = \gamma_0 < \gamma_1 < \dots < \gamma_n < \dots$$

be the increasing sequence of jump times of the process I_t . It is known that

$$\lim_{n \rightarrow \infty} \gamma_n = +\infty \quad \text{almost surely.}$$

Using Eq. (13), if the initial state is $X_0 = e_i$, then the value of the process immediately after the first jump satisfies

$$X_{\gamma_1} = e_i + \sum_{j \in \{1, 2\}} (e_j - e_i) \mathbf{I}_{\Pi_{ij}(F_{\gamma_1})}(I_{\gamma_1}). \quad (14)$$

This means that the process X_t experiences a jump at time γ_1 (i.e., $X_{\gamma_1} \neq X_{\gamma_1-}$) if and only if the value I_{γ_1} lies within the interval $\Pi_{ij}(F_{\gamma_1})$ for some $j \neq i$. If this condition is not met, the process X_t remains unchanged at γ_1 . Hence, the jump times of the process X_t form a subset of the sequence $\{\gamma_1, \gamma_2, \dots\}$.

In the regime-switching models, the behavior of the process depends on the state of the Markov chain, and many pricing formulas or probabilistic analyses involve expectations of the form $\mathbb{E} \left[e^{\int_0^T \eta(X_s) ds} \right]$. If these exponential moments were infinite, the corresponding option prices or risk metrics would be undefined, rendering the model impractical. Moreover, finiteness ensures integrability conditions required for the application of Girsanov transformations and other change-of-measure techniques that are fundamental in risk-neutral pricing frameworks.

Theorem 2.2. Consider the stochastic system defined by (7) and (12) for the pair (F_t, X_t) . Assume that the sequence $(\eta_i)_{i=1,2}$ is nondecreasing, meaning that $\eta_1 \leq \eta_2$. Let

$$\hat{\pi}_{12} := \sup_{u \in \mathbb{R}} \pi_{12}(u), \quad \hat{\pi}_{21} := \inf_{u \in \mathbb{R}} \pi_{21}(u),$$

and

$$\hat{\pi}_1 := -\pi_{11}(u) = \hat{\pi}_{12}, \quad \hat{\pi}_2 := -\pi_{22}(u) = \hat{\pi}_{21}.$$

Suppose

$$\hat{\pi}_{21} > 0, \quad \hat{\pi}_{12} + \hat{\pi}_{21} \leq \pi_{12}(u) + \pi_{21}(u), \quad \text{for all } u \in \mathbb{R}. \quad (15)$$

and

$$\hat{\Pi}_\eta := \begin{pmatrix} -\hat{\pi}_1 + \eta_1 & \hat{\pi}_{12} \\ \hat{\pi}_{21} & -\hat{\pi}_2 + \eta_2 \end{pmatrix}.$$

Define

$$\hat{z} := - \max_{\hat{\eta} \in \text{spec}(\hat{\Pi}_\eta)} \text{Re}(\hat{\eta}).$$

It follows that there is a positive constant A satisfying

$$\mathbb{E} \left[e^{\int_0^t \eta(X_s) ds} \right] \leq A e^{-\hat{2}t}, \quad \text{for all } t > 0.$$

Proof. Define $\hat{\Pi}_{12} = [0, \hat{\pi}_{12}]$, $\hat{\Pi}_{21} = [\hat{\pi}_{12}, \hat{\pi}_{12} + \hat{\pi}_{21}]$, $v(e_1, y) = \mathbf{I}_{\hat{\Pi}_{12}}(y)$, and $v(e_2, y) = -\mathbf{I}_{\hat{\Pi}_{21}}(y)$. Assume that $\hat{X}_t$ be the solution of the following stochastic differential equation

$$d\hat{X}_t = \int_{[0, H] \times [0, H]} v(\hat{X}_{t-}, y) N(dt, dy), \quad \hat{X}_0 = X_0 \quad (16)$$

Consequently, the process $\hat{X}_t$ forms a Markov chain with transition rate matrix given by $(\hat{\pi}_{ij})$. Observe that the process $\{\hat{X}_t\}$ is independent of the element $\omega \in \Omega$, a key fact utilized in the subsequent arguments. Remember that the collection $\{\gamma_k : k \geq 1\}$ represents all jump times of the Poisson point process I . Consequently, both processes X and $\hat{X}_t$ only experience jumps at times within the set $\{\gamma_k : k \geq 1\}$, as follows from their representations in (13) and (16). Therefore, to establish that $X_t \leq \hat{X}_t$ holds almost surely for all $t > 0$, it suffices to prove that almost surely

$$X_{\gamma_k} = e_1 \quad \text{and} \quad \hat{X}_{\gamma_k} = e_2$$

for every $k \geq 1$.

Assuming that $X_0 = \hat{X}_0 = e_1$, we have

$$X_{\gamma_1} = e_1 + \mathbf{I}_{\Pi_{12}(F_{\gamma_1})}(I_{\gamma_1}), \quad \hat{X}_{\gamma_1} = e_1 + \mathbf{I}_{\hat{\Pi}_{12}}(I_{\gamma_1}).$$

From the definitions of $\Pi_{12}(u)$ and $\hat{\Pi}_{12}$, it follows directly that if $I_{\gamma_1} \in \Pi_{12}(F_{\gamma_1})$, then necessarily $I_{\gamma_1} \in \hat{\Pi}_{12}$. Therefore, if $X_{\gamma_1} = e_2$, it follows that $\hat{X}_{\gamma_1} = e_2$ must also hold.

Assuming that $X_0 = \hat{X}_0 = e_2$, we have

$$X_{\gamma_1} = e_2 - \mathbf{I}_{\Pi_{21}(F_{\gamma_1})}(I_{\gamma_1}), \quad \hat{X}_{\gamma_1} = e_2 - \mathbf{I}_{\hat{\Pi}_{21}}(I_{\gamma_1}).$$

Assuming that $\hat{X}_{\gamma_1} = e_1$, we have $\pi_1(\gamma_1) \in \hat{\Pi}_{21}$, and

$$\pi_1(\gamma_1) \leq \hat{\pi}_{12} + \hat{\pi}_{21}, \quad \text{and} \quad \pi_1(\gamma_1) \geq \hat{\pi}_{12} \geq \pi_{12}(F_{\gamma_1}).$$

Using the assumption that $\hat{\pi}_{12} + \hat{\pi}_{21} \leq \pi_{12}(u) + \pi_{21}(u)$ for all $u \in \mathbb{R}$, we conclude that $\pi_1(\gamma_1) \in \Pi_{21}(F_{\gamma_1})$, which implies $X_{\gamma_1} = e_1$. Hence, we have $X_{\gamma_1} = e_1$ and $\hat{X}_{\gamma_1} = e_2$, regardless of whether the initial states satisfy $X_0 = \hat{X}_0 = e_1$ or e_2 . By applying a similar argument, we can also establish that

$$X_{\gamma_{k+1}} = e_1 \quad \text{and} \quad \hat{X}_{\gamma_{k+1}} = e_2 \quad \text{a.s. if} \quad X_{\gamma_k} = \hat{X}_{\gamma_k}, \quad k \geq 2.$$

Now, assuming $X_{\gamma_k} = e_1$ and $\hat{X}_{\gamma_k} = e_2$, we aim to show that $X_{\gamma_{k+1}} = e_1$ and $\hat{X}_{\gamma_{k+1}} = e_2$. Under this assumption, we have

$$X_{\gamma_{k+1}} = e_1 + \mathbf{I}_{\Pi_{12}(F_{\gamma_{k+1}})}(I_{\gamma_{k+1}}),$$

and

$$\hat{X}_{\gamma_{k+1}} = e_2 - \mathbf{I}_{\hat{\Pi}_{21}}(I_{\gamma_{k+1}}).$$

Assuming that $\hat{X}_{\gamma_{k+1}} = e_1$, we have $I_{\gamma_{k+1}} \in \hat{\Pi}_{21}$, and

$$\hat{\pi}_{12} + \hat{\pi}_{21} > I_{\gamma_{k+1}} \geq \hat{\pi}_{12} \geq \pi_{12}(F_{\gamma_{k+1}}).$$

Combining this with the assumption that $\hat{\pi}_{12} + \hat{\pi}_{21} \leq \pi_{12}(u) + \pi_{21}(u)$, we deduce that

$$\pi_{12}(F_{\gamma_{k+1}}) \leq \pi_{12}(F_{\gamma_{k+1}}) + \pi_{21}(F_{\gamma_{k+1}}),$$

this leads to the conclusion that $I_{\gamma_{k+1}} \in \Pi_{21}(F_{\gamma_{k+1}})$ and consequently $X_{\gamma_{k+1}} = e_1 = \hat{X}_{\gamma_{k+1}}$.

If $\hat{X}_{\gamma_{k+1}} = e_2$, it is straightforward to verify that $X_{\gamma_{k+1}} = e_1$ and $\hat{X}_{\gamma_{k+1}} = e_2$. Therefore, we conclude that $X_{\gamma_{k+1}} = e_1$ and $\hat{X}_{\gamma_{k+1}} = e_2$. In summary, we have demonstrated that

$$X_t = e_1, \quad \hat{X}_t = e_2, \quad \text{for all } t > 0.$$

Using the fact that the sequence (η_1, η_2) is nondecreasing, we obtain

$$\eta(X_t) \leq \eta(\hat{X}_t).$$

Therefore,

$$\mathbb{E} \left[e^{\int_0^t \eta(X_s) ds} \right] \leq \mathbb{E} \left[e^{\int_0^t \eta(\hat{X}_s) ds} \right], \quad t > 0.$$

From [54], one can find a constant $A > 0$ so that

$$\mathbb{E} \left[e^{\int_0^t \eta(X_s) ds} \right] \leq \mathbb{E} \left[e^{\int_0^t \eta(\hat{X}_s) ds} \right] \leq A e^{-\hat{2}t}.$$

This completes the proof.

In [Theorem 2.2](#), the construction of the process $\hat{X}_t$ relies on the monotone behavior of the parameters η_1 and η_2 . When $\eta_1 > \eta_2$, to effectively control $\int_0^t \eta(X_s) ds$ associated with X_t through a Markov chain, it is necessary to adjust the definition of the transition rates $(\hat{\pi}_{ij})$ in the following manner

$$\hat{\pi}_{12} = \inf_{u \in \mathbb{R}} \pi_{12}(u), \quad \hat{\pi}_{21} = \sup_{u \in \mathbb{R}} \pi_{21}(u).$$

Under this adjustment, the following inequality remains valid

$$\mathbb{E} \left[e^{\int_0^t \eta(X_s) ds} \right] \leq \mathbb{E} \left[e^{\int_0^t \eta(\hat{X}_s) ds} \right] \leq A e^{-\hat{z}t},$$

where $\hat{z}$ is associated with the matrix $\hat{\Pi}_\eta$ constructed using the modified definitions of $\hat{\pi}_{ij}$ as described above.

2.1. Equivalent probability measure

For pricing options on energy futures, we aim to express the dynamics of these energy futures under a risk-neutral probability measure. The model presented in this study is clearly incomplete for the commodity market. Moreover, we face the challenge of unhedgeable regime switching risk, and the intrinsic inefficiencies associated with energy storage preclude the effective use of the futures process, F_t , for hedging derivatives. Moreover, the inability to effectively store energy leads to significant fluctuations in the spot price of energy, driven by various factors such as weather conditions, supply and demand dynamics, infrastructure constraints, and regulatory changes, among others. Such uncertainties complicate the accurate modeling of the energy spot price process and the development of effective hedging strategies. In traditional financial markets, the assumption of a risk-neutral measure facilitates the construction of a pricing framework where the discounted asset price process behaves as a martingale. However, in the energy markets, this may not hold, because of the unique nature of market dynamics and the presence of unhedgeable risks. Yet, to ensure arbitrage-free pricing and enable consistent option valuation, it remains necessary to define a risk-neutral probability space, even in the presence of market incompleteness. Although perfect replication is not feasible in practice, a mean-self-financing strategy can still be constructed, as discussed in the next section, ensuring no expected gain or loss. That is, on average, the strategy neither generates profit nor incurs a loss.

In order to define a risk neutral measure, we employ Esscher transforms. The use of Esscher transforms in selecting an equivalent martingale measure for option pricing can be traced back to [\[38\]](#). As pointed out by [\[40\]](#), it can be justified as the maximization of the expected utility of economic agents, and it furthermore supports empirical features such as the implied volatility smirk and negative skewness risk premium. The use of Esscher transforms for option pricing in regime switching models has been adopted by several authors such as [\[38–43\]](#), and [\[55\]](#), among others. Here we adapt this approach to the regime switching energy futures dynamics defined by [\(8\)](#) in order to derive an equivalent risk-neutral martingale measure and a pricing formula. However, unlike the references above-mentioned, our model has time-varying volatility. This requires the traditional setting to be slightly extended to deal with this feature.

Recall that $Y_t = \ln(F_t/F_0)$ is a continuous special martingale. Let Θ be the set of $\mathbb{R}^2$ -valued continuous bounded functions with domain $[0, T]$, that is

$$\Theta = \left\{ (\theta_1, \theta_2) \mid \theta_i : [0, T] \rightarrow \mathbb{R} \text{ is continuous, for } i = 1, 2 \right\}.$$

For every $\theta \in \Theta$ we set $\theta_t := \langle \theta(t), X_t \rangle$ where $\theta(t) = (\theta_1(t), \theta_2(t))^T$. Since θ_t is *cadlag*, it is consequently bounded. Thus θ_t is integrable with respect to Y in the sense of stochastic integration. The parameter θ is called the *Esscher transform parameter*. For every $\theta \in \Theta$ we define

$$Y_t^\theta := \int_0^t \theta_s dY_s, \quad L_t^\theta := \int_0^t \frac{1}{2} \sigma^2(s, T, X_s) \theta_s (\theta_s - 1) ds + \int_0^t \theta_s \sigma(s, T, X_s) dB_s.$$

Then, Ito's Lemma gives

$$\exp(Y_t^\theta) = 1 + \int_0^t \exp(Y_s^\theta) dL_s^\theta.$$

Therefore L_t^θ is the logarithmic transform of Y_t . Now the predictable finite-variation part of L_t^θ is

$$M_t^\theta := \int_0^t \frac{1}{2} \sigma^2(s, T, X_s) \theta_s (\theta_s - 1) ds,$$

Hence, M_t^θ is the Laplace cumulant process of Y_t^θ (for details on the Laplace cumulant process, refer to [Appendix A](#)). Elementary calculus yields

$$\exp(M_t^\theta) = 1 + \int_0^t \exp(M_s^\theta) dM_s^\theta.$$

As a result, $\widetilde{M}_t^\theta := M_t^\theta$ is the exponential transform of itself. Now, define

$$Z_T^\theta = \exp \left(\int_0^T \theta_s dY_s - \widetilde{M}_T^\theta \right) = \exp \left(\int_0^T \theta_s \sigma(s, T, X_s) dB_s - \frac{1}{2} \int_0^T \theta_s^2 \sigma^2(s, T, X_s) ds \right).$$

Using Theorem 2.2, we obtain exponential integrability of the functional $\int_0^T \theta_s^2 \sigma^2(s, T, X_s) ds$, which verifies Novikov's condition. As a result, Z_T^θ is a true martingale, and the measure change via Girsanov's theorem is well-defined. Then, the Esscher transformed equivalent martingale measure, $\mathbb{P}^\theta$, is defined by the Radon-Nykodym derivative

$$\frac{d\mathbb{P}^\theta}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = Z_T^\theta.$$

We want to choose θ in order to determine a risk-neutral probability measure. From Girsanov Theorem we know that

$$B_t^\theta = B_t - \int_0^t \theta_s \sigma(s, T, X_s) ds$$

is a standard Brownian motion under $\mathbb{P}^\theta$. Therefore, the differential, dF_t of the price process F_t satisfies

$$dF_t = \theta_t \sigma^2(t, T, X_t) F_t dt + \sigma(t, T, X_t) F_t dB_t^\theta.$$

Under the risk-neutral measure, the non-discounted price of an energy futures must follow a martingale process. Specifically, in a risk-neutral world, all tradable assets are expected to grow at the risk-free rate when discounted, meaning that their discounted price processes must be martingales. Since the futures price is typically quoted in non-discounted terms and does not require upfront investment, its expected future value under the risk-neutral measure must equal its current price to prevent arbitrage opportunities. Therefore, the non-discounted futures price must be a martingale under the risk-neutral measure, ensuring that its expected change over time, conditioned on current information, is zero.

It becomes evident that F_t satisfies the martingale property if and only if $\theta_t = 0$. This is the case where $\theta^* = (\theta_1^*(t), \theta_2^*(t)) \in \Theta$ must vanish. On the other hand, the fundamental theorem of asset pricing states that the regime switching energy futures dynamics defined by (8) admits no-arbitrage opportunities if and only if the Esscher transform parameter is θ^* . Recall that $\xi_t = \langle \bar{\xi}, X_t \rangle$. For simplicity, we set

$$\mathbb{P}^* := \mathbb{P}^{\theta^*}, \quad B_t^* := B_t^{\theta^*}, \quad \sigma_t = \sqrt{\frac{1 - e^{-2\kappa(T-t)}}{2\kappa(T-t)}} \xi_t,$$

and we refer to $\mathbb{P}^*$ as the risk-neutral probability measure. Moreover, $B_t^* = B_t$, and the futures price dynamics under the risk-neutral probability measure $\mathbb{P}^*$ is expressed as follows

$$dF_t = \sigma_t F_t dB_t^*, \quad F_0 = f > 0. \quad (17)$$

Since the parameter σ_t is random, when viewed under the risk-neutral measure the model can be seen as a Black–Scholes model with parameters that vary over time in a random environment.

3. Option pricing model

In terms of transitioning between different states, the rate at which the desired phenomenon leaves state e_i will be denoted as λ_i , and the time it takes to leave state e_i will be denoted as τ_i . It is assumed that

$$\mathbb{P}(\tau_i > t) = e^{-\lambda_i(X_i)t}, \quad i = 1, 2.$$

The waiting time for switching between regimes remains unchanged under the measure change, meaning that the distribution of τ_i is unaffected by the transition from $\mathbb{P}$ to $\mathbb{P}^*$. The closed form formula for option prices relies heavily on the assumption of exponential holding time. This assumption is crucial in deriving the formula. The closed form formula not only makes it easier to calculate option prices but also provides a valuable insight into how various components of the model influence option prices. The memoryless property is a well-known result of the exponential holding time. The exponential holding time's mathematical tractability, coupled with its memoryless property, makes it a practical choice. If we do not have a specific theory regarding the distribution of τ_i , we can adopt a stance of uniform ignorance. This means that regardless of the length of time spent in the current state, we hold an equal level of uncertainty about when the next state change will occur.

The classical Black–Scholes model has been successful, but there has been a recent surge in attention towards some empirical phenomena. The Black–Scholes model relies on two key assumptions: firstly, that the underlying asset distribution follows a lognormal pattern, and secondly, that the volatility remains constant. On the other hand, empirical evidence indicates that asset distribution shows characteristics of leptokurtosis and asymmetry, and volatility demonstrates clustering behavior. Model (17) exhibits asymmetric leptokurtic characteristics, negative skewness, and negative correlation with future volatility by incorporating a hidden Markov process into the Black–Scholes exponential Brownian motion. In this hidden Markov process, the drift and volatility parameters vary based on the state, as discussed by [46]. In their study, [47] demonstrated that the Markov switching model outperforms the Black–Scholes model in capturing various characteristics of futures returns, such as volatility clustering, leverage effects, excess volatility, and expected returns.

For the proposed energy futures model (17), the Poisson process governs the transitions between the states of the Markov chain. Specifically, the time spent in state e_i before transitioning to state e_j , denoted τ_{ij} , follows an exponential distribution with rate π_{ij} , such that

$$\mathbb{P}(\tau_{ij} > t) = e^{-\pi_{ij}t}, \quad \text{for } i, j = 1, 2 \text{ and } i \neq j.$$

The transition times τ_{ij} are unaffected by the change of measure from $\mathbb{P}$ to $\mathbb{P}^*$, as the measure change impacts the dynamics of the continuous component rather than the Markov chain itself. The matrix $\Pi = (\pi_{ij})$ defines the transition rates of the hidden Markov chain, where π_{ij} represents the rate of transitioning from state e_i to state e_j . Evidently, the proposed model reduces to the Black–Scholes model when both transition rates, π_{12} and π_{21} , are set to zero. In this case, the Markov chain ceases to switch between states, and the model describes a single-regime framework consistent with Black–Scholes dynamics.

When the volatility and risk-free rates are constant in each regime, a simple and efficient semi-closed form formula for the price of a European call option can be obtained, as in [56] for instance. However, this is not possible here, because the volatility is time-varying in each model. In order to establish our formula, we apply the celebrated Fourier transform technique to our regime switching model, a method introduced by [57]. Additional examples of applying the Fourier transform technique in regime switching models, beyond this article, can be found in [41,58], among others.

Theorem 3.1. For a European call option with a strike price of K and maturity T , the price $U(f, x, t)$ at any time $0 \leq t < T$, given the current energy futures price $F_t = f > 0$ and regime state $X_t = x \in \{e_1, e_2\}$, is determined under the risk-neutral probability measure $\mathbb{P}^*$ as follows:

$$U(f, x, t) = \mathbb{E}^* \left[e^{-\int_t^T r_s ds} (F_T - K)_+ \right],$$

where r_t is the market interest rate modulated by the hidden Markov chain X defined by

$$r_t := \langle \bar{r}, X_t \rangle, \quad \bar{r} = (r_1, r_2)^\top \in \mathbb{R}^2.$$

Then,

$$U(f, x, t) = \frac{f e^{-\alpha k}}{\pi} \int_0^\infty e^{-ivk} \psi(v) dv, \quad (18)$$

with

$$\psi(v) = \frac{\left\langle x \exp \left\{ \int_t^T (\text{diag}(g(s, v - i(\alpha + 1)))) ds + \Pi(T - t) \right\}, \mathbf{1} \right\rangle}{(1 + \alpha + iv)(\alpha + iv)},$$

where

$$\begin{aligned} k &= \ln(K/f), \quad \mathbf{1} = (1, 1)^\top, \\ g(s, v) &= (g_1(s, v), g_2(s, v))^\top, \quad g_j(s, v) = -r_j - \frac{1}{2} i v \sigma_j^2(s) - \frac{1}{2} v^2 \sigma_j^2(s), \\ \sigma_j(s) &= \xi_j \sqrt{\frac{1 - e^{-2\kappa(T-s)}}{2\kappa(T-s)}}. \end{aligned}$$

Proof. Define $Y_T = \ln(F_T/f)$. Let $\mathcal{F}_t^X$ denotes the subfiltration generated by the hidden Markov chain X over the interval $[t, T]$. In this proof, $\mathbb{P}^*$ indicates the risk-neutral measure conditioned on $F_t = f$ and $X_t = x$. The symbol $\mathbb{E}^*$ refers to the expectation taken under $\mathbb{P}^*$. Assuming that $F_{Y_T | \mathcal{F}_T^X}(v)$ is the conditional distribution of Y_T given $\mathcal{F}_T^X$ under $\mathbb{P}^*$, we denote the conditional characteristic function of Y_T given $\mathcal{F}_T^X$ by

$$\varphi_{Y_T | \mathcal{F}_T^X}(v) := \mathbb{E}^* [e^{ivY_T} | \mathcal{F}_T^X].$$

In this case, Theorem 2.2 ensures $\varphi_{Y_T | \mathcal{F}_T^X}(v)$ takes finite values, and we have

$$(F_T - K)_+ = f(e^{Y_T} - \frac{K}{f})_+ = f(e^{Y_T} - e^k)_+,$$

where $k = \ln(K/f)$. Therefore

$$U(f, x, t) = \mathbb{E}^* \left[e^{-\int_t^T r_s ds} (F_T - K)_+ \right] = f C(k),$$

where

$$C(k) = \mathbb{E}^* \left[e^{-\int_t^T r_s ds} (e^{Y_T} - e^k)_+ \right].$$

The Fourier transform, $\psi(v)$, of the dampened call option price, $e^{\alpha k} C(k)$, with dampening parameter $\alpha > 0$, can be calculated as follows:

$$\begin{aligned} \psi(v) &= \int_{\mathbb{R}} e^{ivk} e^{\alpha k} C(k) dk \\ &= \mathbb{E}^* \left[e^{-\int_t^T r_s ds} \int_{\mathbb{R}} e^{ivk} e^{\alpha k} E^* [(e^{Y_T} - e^k)_+ | \mathcal{F}_T^X] dk \right] \\ &= \mathbb{E}^* \left[e^{-\int_t^T r_s ds} \int_{\mathbb{R}} e^{ivk} \left(\int_k^\infty (e^{s+\alpha k} - e^{(1+\alpha)k}) F_{Y_T | \mathcal{F}_T^X}(ds) \right) dk \right] \end{aligned}$$

$$= \mathbb{E}^* \left[e^{-\int_t^T r_s ds} \int_{\mathbb{R}} \left(e^{s+\alpha k} - e^{(1+\alpha)k} \right) e^{ivk} dk \right] F_{Y_T | \mathcal{F}_T^X}(ds) \Big].$$

Using

$$\int_{-\infty}^s (e^{s+\alpha k} - e^{(1+\alpha)k}) e^{ivk} dk = \frac{e^{(1+\alpha+iv)s}}{\alpha+iv} - \frac{e^{(1+\alpha+iv)s}}{1+\alpha+iv} = \frac{e^{i(v-(\alpha+1))s}}{(1+\alpha+iv)(\alpha+iv)},$$

we obtain

$$\psi(v) = \mathbb{E}^* \left[e^{-\int_t^T r_s ds} \int_{\mathbb{R}} \frac{e^{i(v-(\alpha+1))s}}{(1+\alpha+iv)(\alpha+iv)} F_{Y_T | \mathcal{F}_T^X}(ds) \right],$$

and thus

$$\psi(v) = \frac{\mathbb{E}^* \left[e^{-\int_t^T r_s ds} \varphi_{Y_T | \mathcal{F}_T^X}(v - i(\alpha+1)) \right]}{(1+\alpha+iv)(\alpha+iv)}.$$

Now

$$dY_t = -\frac{1}{2} \sigma_t^2 dt + \sigma_t dB_t^*.$$

Conditioned on $\mathcal{F}_T^X$, this is a mere drifted Brownian motion. Hence

$$\begin{aligned} \varphi_{Y_T | \mathcal{F}_T^X}(v) &= \exp \left(iv \int_t^T -\frac{1}{2} \sigma_s^2 ds - \frac{1}{2} v^2 \int_t^T \sigma_s^2 ds \right), \\ e^{-\int_t^T r_s ds} \varphi_{Y_T | \mathcal{F}_T^X}(v) &= \exp \left(\int_t^T \left(-r_s - \frac{1}{2} iv \sigma_s^2 - \frac{1}{2} v^2 \sigma_s^2 \right) ds \right) \\ &= \exp \left(\int_t^T \langle g(s, v), X_s \rangle ds \right). \end{aligned}$$

Taking the expectation yields

$$\begin{aligned} \mathbb{E}^* \left[e^{-\int_t^T r_s ds} \varphi_{Y_T | \mathcal{F}_T^X}(v) \right] &= \mathbb{E}^* \left[\exp \left(\int_t^T \langle g(s, v), X_s \rangle ds \right) \right] \\ &= \left\langle x \exp \left\{ \int_t^T (\text{diag}(g(s, v))) ds + \Pi(T-t) \right\}, \mathbf{1} \right\rangle. \end{aligned}$$

Substituting v with $v - i(\alpha+1)$, we obtain the expression for $\psi(v)$ stated in the theorem. Then, using the inverse transform, we get

$$e^{\alpha k} C(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ivk} \psi(v) dv.$$

However, because $e^{\alpha k} C(k)$ is a real-valued function, its Fourier transform, $\psi(v)$, exhibits the following symmetries: the real part of $\psi(v)$ is an even function, and the imaginary part of $\psi(v)$ is an odd function. It follows that

$$C(k) = \frac{e^{-\alpha k}}{\pi} \int_0^{\infty} e^{ivk} \psi(v) dv.$$

As a result, we have

$$U(f, x, t) = f \frac{e^{-\alpha k}}{\pi} \int_0^{\infty} e^{ivk} \psi(v) dv,$$

as desired.

Remark 1. When there are two regimes and the true regime state is uncertain, we can utilize a probability distribution $(\tilde{\alpha}, 1 - \tilde{\alpha})$, where $0 \leq \tilde{\alpha} \leq 1$, to calculate the option price. This is done according to the formula $U = \tilde{\alpha} U(Y_0, e_1, 0) + (1 - \tilde{\alpha}) U(Y_0, e_2, 0)$. Here $U(Y_0, e_1, 0)$ and $U(Y_0, e_2, 0)$ are the prices of European call options in regimes 1 and 2, respectively. Moreover, the integral involved in option pricing model can be obtained based on the Gauss quadrature.

It is easy to see that the option prices under the regime switching model for contracts with short durations are close to those given by the Black–Scholes models. In other words, the option prices calculated using Eq. (18) fall within the range of prices determined by the two corresponding classical Black–Scholes models. Moreover, as the contract duration increases, the initial state's significance diminishes, as the Markov chain converges to its stationary distribution.

3.1. Hedging strategy

From [59,60], the incompleteness of model (17) can be attributed to the fact that the futures price is influenced not only by the Wiener process B but also by the hidden Markov process X_t . This means that there is more than one equivalent risk-neutral martingale measure. Different risk-neutral measures will yield different replication strategies. Furthermore, there may not even be a self-financing replication strategy for each contingent claim. One way to deal with this situation was given by [61] who define

the concept of *mean-self-financing* replication strategies. For such strategies, the replication is not self-financing but the expected cost/profit resulting from maintaining the replication is null. Then, an *optimal* mean-self-financing replication strategy is found under the so-called *minimal martingale measure*, for which the expectation of square of the replication cost is minimized. In [42], it is shown that the risk-neutral measure $\mathbb{P}^*$ with Esscher transform parameter θ^* is the minimal martingale measure and thus provides the optimal mean-self-financing replication strategy.

We seek a hedging portfolio consisting of energy futures contracts F with maturity T and an amount of money invested in the money market. The futures price dynamics are described by the regime switching model (17), and any amount B_t invested in the money market satisfies the dynamic $dB_t = B_t r_t dt$, where r_t is the short-term interest rate. Let $U(F_t, X_t, t)$ denotes the price at time t of a European call option with maturity T and strike K . To hedge this contingent claim, the portfolio holds γ_t units of the energy futures F_t and an amount β_t in the money market, where γ_t and β_t are real-valued, adapted processes. Such a portfolio is called a mean-self-financing replicating portfolio if there exists a martingale M_t such that

$$dU(F_t, X_t, t) = \gamma_t dF_t + \beta_t r_t dt + dM_t,$$

and

$$U(F_t, X_t, t) = \gamma_t F_t + \beta_t \quad \text{for every } 0 \leq t \leq T.$$

Theorem 3.2. Let $U(F, X, t)$ be the European option price written on energy futures at time t , when $F_t = F$ and $X_t = X$. Then $U(F, X, t)$ satisfies the Cauchy problem

$$AU(F, X, t) - r_t U(F, X, t) + \left\langle \bar{U}(F, t), \Pi X \right\rangle = 0, \quad (19)$$

where

$$AU(F, X, t) = \left(\frac{\partial}{\partial t} + r_t F \frac{\partial}{\partial F} + \frac{1}{2} \sigma_t^2 F^2 \frac{\partial^2}{\partial F^2} \right) U(F, X, t)$$

and

$$\bar{U}(F, t) = (U(F, e_1, t), U(F, e_2, t)).$$

Moreover, a mean-self-financing replicating portfolio in $(\Omega, \mathcal{F}, \mathbb{P}^*)$ is given by

$$\gamma_t = \frac{\partial U}{\partial F}(F_t, X_t, t) \quad \text{and} \quad \beta_t = U(F_t, X_t, t) - \frac{\partial U}{\partial F}(F_t, X_t, t) F_t.$$

Proof. By the fundamental theorem of asset pricing, under the risk-neutral measure $\mathbb{P}^*$, the discounted option price process $Y_t = e^{-\int_0^t r_s ds} U(F_t, X_t, t)$ is a martingale, as the discounted price of a security derivative equals its conditional expected payoff at time t , ensuring no arbitrage in the market. Recall that X_t has the semi-martingale representation

$$dX_t = \Pi X_t dt + dV_t,$$

where Π is the generator matrix of the Markov chain governing X_t , and V_t is a martingale in $(\Omega, \mathcal{F}, \mathbb{P}^*)$. Using Ito's Lemma, the total differential of Y_t is

$$\begin{aligned} dY_t = & e^{-\int_0^t r_s ds} \left[AU(F_t, X_t, t) - r_t U(F_t, X_t, t) + \left\langle \bar{U}(F_t, t), \Pi X_t \right\rangle \right] dt \\ & + e^{-\int_0^t r_s ds} \sigma_t F_t \frac{\partial U}{\partial F} dF_t + e^{-\int_0^t r_s ds} \left\langle \bar{U}(F_t, t), dV_t \right\rangle. \end{aligned}$$

The martingale property of Y_t implies that the drift term (the coefficient of dt) must vanish, resulting in the partial differential equation

$$AU(F, X, t) + \left\langle \bar{U}(F, t), \Pi X \right\rangle - r_t U(F, X, t) = 0,$$

which establishes that $U(F, X, t)$ satisfies the Cauchy problem given in (19).

Since $U(F_t, X_t, t) = e^{\int_0^t r_s ds} Y_t$, applying Ito's Lemma gives

$$dU(F_t, X_t, t) = r_t U(F_t, X_t, t) dt + e^{\int_0^t r_s ds} dY_t.$$

If $\beta_t := U(F_t, X_t, t) - \frac{\partial U}{\partial F}(F_t, X_t, t) F_t$ represents the amount held in the money market, then its growth is described by

$$d\beta_t = \beta_t r_t dt = r_t U(F_t, X_t, t) dt - r_t \frac{\partial U}{\partial F}(F_t, X_t, t) F_t dt.$$

With $\gamma_t := \frac{\partial U}{\partial F}(F_t, X_t, t)$ units of the energy futures F_t held in the same portfolio, we have

$$\gamma_t dF_t + \beta_t r_t dt = \frac{\partial U}{\partial F}(F_t, X_t, t) dF_t + r_t U(F_t, X_t, t) dt - r_t \frac{\partial U}{\partial F}(F_t, X_t, t) F_t dt$$

$$\begin{aligned}
&= \frac{\partial U}{\partial F} (F_t, X_t, t) \sigma_t F_t dW_t + dU (F_t, X_t, t) - e^{\int_0^t r_s ds} dY_t \\
&= dU (F_t, X_t, t) + dM_t,
\end{aligned}$$

where the martingale M_t satisfies

$$dM_t = \frac{\partial U}{\partial F} (F_t, X_t, t) \sigma_t F_t dW_t - e^{\int_0^t r_s ds} dY_t.$$

Since $\gamma_t F_t + \beta_t = U (F_t, X_t, t)$, this identifies a mean-self-financing replicating portfolio.

4. Estimation of parameters

The EM algorithm, short for expectation–maximization, was introduced by [62] and further developed by [30]. It is a highly effective machine learning technique used to find the maximum likelihood estimates of parameters in probabilistic models involving latent variables or incomplete data. On the other hand, the maximum likelihood estimation method involves estimating parameters based on visible data. The EM algorithm operates through two main iterative steps: the Expectation (E) step and the Maximization (M) step. In the E-step, the algorithm deals with the expectation of the log-likelihood function. This step is concerned with determining the conditional distribution of the hidden variables, based on the observed data and the current parameter estimates. Subsequently, in the M-step, the parameters of the model are revised in order to maximize this expectation. These steps are repeated iteratively until the parameter updates fall below a pre-set threshold, indicating convergence. The EM algorithm is commonly used for parameter estimation in regime switching Markov models and other hidden Markov models (e.g., [63]). However, its versatility extends to other models as well. For instance, it has been applied to parameter estimation in Gaussian mixture and hidden Markov models (e.g., [64]). Additionally, [65] discussed the use of the EM algorithm for parameter estimation in dependence tree models, while [66] demonstrated its application in nonparametric estimation of mixing distributions.

Let $\mathbb{F} = (F_{t_0}, \dots, F_{t_n})$ be a vector of observed futures prices from our energy futures dynamics, and let $\mathbb{X} = (X_{t_0}, \dots, X_{t_n})$ be possible values for the hidden regime X_t . The purpose of the EM algorithm is to determine the unknown parameter vector ζ of the distribution of Markov chain X_t that maximizes the probability of observing $\mathbb{F}$. Obviously, this is the same as maximizing the log-likelihood function, $\mathcal{L}(\zeta)$, defined as

$$\mathcal{L}(\zeta) = \log(\mathbb{P}(\mathbb{F} | \zeta)).$$

The EM algorithm employs an iterative process to maximize $\mathcal{L}(\zeta)$. The goal of iteration $m + 1$ is to iteratively improve

$$\mathcal{L}(\zeta) - \mathcal{L}(\zeta^{(m)}) = \log(\mathbb{P}(\mathbb{F} | \zeta)) - \log(\mathbb{P}(\mathbb{F} | \zeta^{(m)})). \quad (20)$$

Now, $\mathbb{X}$ can take finitely many values x so that

$$\mathbb{P}(\mathbb{F} | \zeta) = \sum_x \mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta) = \sum_x \frac{\mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta)}{\mathbb{P}(x | \mathbb{F}, \zeta^{(m)})} \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)}).$$

Hence,

$$\mathcal{L}(\zeta) - \mathcal{L}(\zeta^{(m)}) = \log \left(\sum_x \frac{\mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta)}{\mathbb{P}(x | \mathbb{F}, \zeta^{(m)})} \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)}) \right) - \log(\mathbb{P}(\mathbb{F} | \zeta^{(m)})).$$

Since the logarithm is a concave function and $x \mapsto \mathbb{P}(x | \mathbb{F}, \zeta^{(m)})$ defines a discrete probability, applying Jensen's inequality for concave functions gives

$$\begin{aligned}
\mathcal{L}(\zeta) - \mathcal{L}(\zeta^{(m)}) &\geq \sum_x \log \left(\frac{\mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta)}{\mathbb{P}(x | \mathbb{F}, \zeta^{(m)})} \right) \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)}) - \log(\mathbb{P}(\mathbb{F} | \zeta^{(m)})) \\
&= \sum_x \log \left(\frac{\mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta)}{\mathbb{P}(x | \mathbb{F}, \zeta^{(m)}) \mathbb{P}(\mathbb{F} | \zeta^{(m)})} \right) \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)})
\end{aligned}$$

With the definition

$$\ell(\zeta | \zeta^{(m)}) := \mathcal{L}(\zeta^{(m)}) + \sum_x \log \left(\frac{\mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta)}{\mathbb{P}(x | \mathbb{F}, \zeta^{(m)}) \mathbb{P}(\mathbb{F} | \zeta^{(m)})} \right) \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)}),$$

the above inequality simplifies to $\mathcal{L}(\zeta) \geq \ell(\zeta | \zeta^{(m)})$. Replacing ζ by $\zeta^{(m)}$ we find that $\mathcal{L}(\zeta^{(m)}) = \ell(\zeta^{(m)} | \zeta^{(m)})$. This suggests that improving the modified likelihood function $\ell(\zeta | \zeta^{(m)})$ also improves the log-likelihood function $\mathcal{L}(\zeta)$. Indeed, it is shown in [67] that the sequence of values $\{\zeta^{(m)}\}$ obtained as $\zeta^{(m+1)} = \arg \max_{\zeta} (\ell(\zeta | \zeta^{(m)}))$ converges to a stationary point of $\mathcal{L}(\zeta)$. Exploiting the basic property that $\log(a/b) = \log(a) - \log(b)$ when $a, b > 0$, we have

$$\begin{aligned}
\ell(\zeta | \zeta^{(m)}) &= \sum_x \log (\mathbb{P}(\mathbb{F} | x, \zeta) \mathbb{P}(x | \zeta)) \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)}) + G(\zeta^{(m)}) \\
&= \sum_x \log (\mathbb{P}(\mathbb{F}, x | \zeta)) \cdot \mathbb{P}(x | \mathbb{F}, \zeta^{(m)}) + G(\zeta^{(m)}) \\
&= \mathbb{E}_{\mathbb{X} | \mathbb{F}, \zeta^{(m)}} [\log (\mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta))] + G(\zeta^{(m)}),
\end{aligned}$$

where $G(\zeta^{(m)})$ is an expression that does not depend on ζ in any manner. Thus

$$\zeta^{(m+1)} = \arg \max_{\zeta} \ell(\zeta | \zeta^{(m)}) = \arg \max_{\zeta} \left\{ \mathbb{E}_{\mathbb{X}|\mathbb{F}, \zeta^{(m)}} [\log(\mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta))] \right\}.$$

The $(m+1)$ -th iteration of the EM algorithm consists of the following two steps:

- (1) E-step: Evaluate $\mathbb{E}_{\mathbb{X}|\mathbb{F}, \zeta^{(m)}} [\log(\mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta))]$.
- (2) M-step: Maximize $\mathbb{E}_{\mathbb{X}|\mathbb{F}, \zeta^{(m)}} [\log(\mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta))]$ with respect to ζ .

We now describe the E-step in the context of our energy futures dynamic. Consider an equidistant partition of the interval $[0, T]$, denoted as $\{0 = t_0 < t_1 < \dots < t_n = T\}$, with a uniform mesh $\delta := t_j - t_{j-1}$. Before we estimate the parameters of the regime switching model described in Eq. (7), we analyze its discrete form using the Euler method as follows:

$$F_{t_j} = F_{t_{j-1}} + \sigma(t_{j-1}, T, X_{t_{j-1}}) F_{t_{j-1}} \sqrt{\delta} \varepsilon_j,$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent standard normal random variables. Clearly, the conditional distribution of F_{t_j} , given $F_{t_{j-1}}$ and X_{t_j} is normally distributed with mean $F_{t_{j-1}}$ and variance $\sigma^2(t_{j-1}, T, X_{t_j}) F_{t_{j-1}}^2 \delta$. With two states in the Markov chain, the conditional density function of F_{t_j} , given $F_{t_{j-1}}$ and X_{t_j} , is

$$G(F_{t_j} | F_{t_{j-1}}, X_{t_j} = e_k, \zeta) = \frac{1}{\sqrt{2\pi\sigma_{k,t_j}^2 F_{t_{j-1}}^2 \delta}} \exp \left\{ -\frac{(F_{t_j} - F_{t_{j-1}})^2}{2\sigma_{k,t_j}^2 F_{t_{j-1}}^2 \delta} \right\}, \quad k = 1, 2,$$

where $\zeta = (\kappa, \xi_1, \xi_2)$, and

$$\sigma_{k,t_j} = \xi_k \sqrt{\frac{1 - e^{-2\kappa(T-t_j)}}{2\kappa(T-t_j)}}, \quad k = 1, 2, \quad j = 1, \dots, n.$$

The following result ties the functions G to the conditional expectation that needs to be evaluated and maximized in the EM algorithm.

Theorem 4.1. *In the E-step of the EM algorithm, the conditional expectation $\mathbb{E}_{\mathbb{X}|\mathbb{F}, \zeta^{(m)}} [\ln \mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta)]$ can be computed as*

$$\begin{aligned} \mathbb{E}_{\mathbb{X}|\mathbb{F}, \zeta^{(m)}} [\ln \mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta)] &= \sum_{k=1}^2 \sum_{j=1}^n \mathbb{P}(X_{t_j} = e_k | \mathbb{F}, \zeta^{(m)}) \\ &\quad \times \ln G(F_{t_j} | F_{t_{j-1}}, X_{t_j} = e_k, \zeta). \end{aligned}$$

Proof. See [13].

As we see, the calculation and maximization of $\mathbb{E}_{\mathbb{X}|\mathbb{F}, \zeta^{(m)}} [\ln \mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta)]$ require the computation of $\mathbb{P}(X_{t_j} = e_k | \mathbb{F}, \zeta^{(m)})$. This involves a two-step process known as *filtering inference* and *smoothing inference*. The filtering inference calculates the probability of the hidden state X_{t_j} at time t_j , given the observed energy futures price F_{t_j} . Specifically, it computes $\mathbb{P}(X_{t_j} = e_k | F_{t_j}, \zeta^{(m)})$. In contrast, the smoothing inference calculates the probability of the hidden state X_{t_j} at time t_j , given all observed energy futures prices $\mathbb{F}$. This corresponds to computing $\mathbb{P}(X_{t_j} = e_k | \mathbb{F}, \zeta^{(m)})$. As detailed by [30], this two-step process can be formulated as follows:

- (1) **Filtering inference:** For each j from 1 to n , compute the probability:

$$\mathbb{P}(X_{t_j} = e_k | F_{t_j}, \zeta^{(m)}) = \frac{\mathbb{P}(X_{t_j} = e_k | F_{t_{j-1}}, \zeta^{(m)}) G(F_{t_j} | X_{t_j} = e_k, F_{t_{j-1}}, \zeta^{(m)})}{\sum_{i=1}^2 [\mathbb{P}(X_{t_j} = e_i | F_{t_{j-1}}, \zeta^{(m)}) G(F_{t_j} | X_{t_j} = e_i, F_{t_{j-1}}, \zeta^{(m)})]},$$

where

$$\mathbb{P}(X_{t_j} = e_k | F_{t_{j-1}}, \zeta^{(m)}) = \sum_{i=1}^2 p_{ki}^{(m)} \mathbb{P}(X_{t_{j-1}} = e_i | F_{t_{j-1}}, \zeta^{(m)}).$$

- (2) **Smoothing inference:** For $j = n-1, \dots, 1$, compute the probability:

$$\mathbb{P}(X_{t_j} = e_k | \mathbb{F}, \zeta^{(m)}) = \sum_{i=1}^2 \left[\frac{\mathbb{P}(X_{t_{j+1}} = e_i | \mathbb{F}, \zeta^{(m)}) \mathbb{P}(X_{t_j} = e_k | F_{t_j}, \zeta^{(m)}) p_{ik}^{(m)}}{\mathbb{P}(X_{t_{j+1}} = e_i | F_{t_j}, \zeta^{(m)})} \right].$$

At the onset of the algorithm, at step $m = 0$, an arbitrary transition probability matrix $(p_{ki}^{(0)})$ for the hidden Markov chain X needs to be provided, along with the probabilities $h_k^{(0)} = \mathbb{P}(X_{t_1} = e_k | F_{t_0}, \zeta^{(0)})$, representing the probability of the state $X_{t_1} = e_k$ given the initial observation F_{t_0} . At step $m+1$, these quantities are updated as follows:

$$h_k^{(m+1)} = \mathbb{P}(X_{t_1} = e_k | \mathbb{F}, \zeta^{(m)}),$$

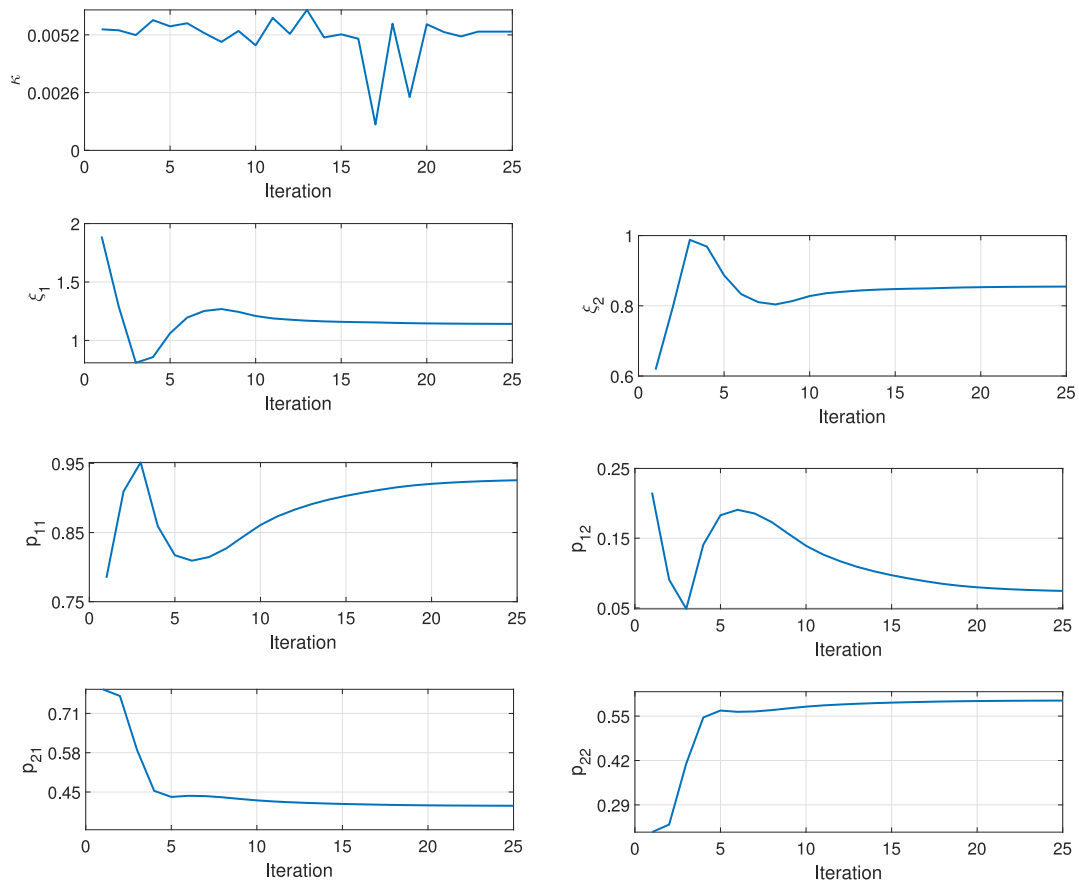


Fig. 1. Parameters estimated by EM algorithm with various iterations.

$$p_{ki}^{(m+1)} = \frac{\sum_{j=1}^{n-1} \mathbb{P}(X_{t_{j+1}} = e_k | \mathbb{F}, \zeta^{(m)}) \frac{p_{ki}^{(m)} \mathbb{P}(X_{t_j} = e_i | F_{t_j}, \zeta^{(m)})}{\mathbb{P}(X_{t_{j+1}} = e_k | F_{t_j}, \zeta^{(m)})}}{\sum_{j=1}^{n-1} \mathbb{P}(X_{t_j} = e_i | \mathbb{F}, \zeta^{(m)})},$$

where $i, k = 1, 2$.

We see that the E-step is concerned with solving for the parameters of the hidden regime switching model. Indeed, after the m th iteration, if we use $p_{ki}^{(m)}$ as estimates for the limiting transition probabilities p_{ki} , then the components π_{ki} can be found from the equality $p_{ki} = (e^{\Pi \delta})_{ki}$. Moreover, if $h_j^{(m)}$ is used as an estimate of $h_j = \mathbb{P}(X_{t_1} = e_j)$, then the values of $\mathbb{P}(X_0 = e_1)$, and $\mathbb{P}(X_0 = e_2)$ can be found by solving the system of linear equations

$$\begin{aligned} h_1 &= p_{11} \mathbb{P}(X_0 = e_1) + p_{12} \mathbb{P}(X_0 = e_2) \\ h_2 &= p_{21} \mathbb{P}(X_0 = e_1) + p_{22} \mathbb{P}(X_0 = e_2). \end{aligned}$$

Now we turn our attention to the M-step of the EM algorithm. At the $(m+1)$ th iteration, the parameters κ and ξ_i (for $i = 1, 2$) are estimated by solving the following maximization problem:

$$\max_{\zeta^{(m+1)}} \mathbb{E}_{\mathbb{X} | \mathbb{F}, \zeta^{(m)}} \left[\ln \mathbb{P}(\mathbb{F}, \mathbb{X} | \zeta) \right].$$

We use the genetic optimization algorithm (GA) to maximize the log-likelihood function, more specifically, to maximize

$$\sum_{j=1}^n \mathbb{P}(X_{t_j} = e_k | \mathbb{F}, \zeta^{(m)}) \times \ln G(F_{t_j} | F_{t_{j-1}}, X_{t_j} = e_k, \zeta)$$

In doing so, if at time t_j the probability $\mathbb{P}(X_{t_j} = e_2 | \mathbb{F})$ surpasses $\mathbb{P}(X_{t_j} = e_1 | \mathbb{F})$, the regime is classified as being in state e_2 ; in the opposite scenario, it is classified as being in state e_1 . We used daily data with a granularity of $\delta = 0.0455$, sourced from the United States gas futures prices spanning from January 2017 to January 2023.

Table 1

Parameters of the developed regime switching energy futures' dynamics obtained using the EM algorithm.

$\kappa = 0.0054$	$p_{11} = 0.9236$	$\pi_{11} = -2.3055$
$\xi_1 = 1.1275$	$p_{12} = 0.0764$	$\pi_{12} = 2.3055$
$\xi_2 = 0.8700$	$p_{21} = 0.4131$	$\pi_{21} = 12.4695$
$\lambda_1 = 2.3055$	$p_{22} = 0.5869$	$\pi_{22} = -12.4695$
$\lambda_2 = 12.4695$	$\mathbb{P}(X_0 = e_1) = 0.4131$	$\mathbb{P}(X_0 = e_2) = 0.5869$

The generator Π determines how the states of the Markov chain either transition or remain constant daily, governed by the one-day transition probability matrix, $P = e^{\Pi\delta}$. The model dynamics highlight significant persistence within each regime, for which distinct economic interpretations can be attributed: state e_1 characterizes a period of high-volatility, while state e_2 is associated with a period of low-volatility. These regimes effectively differentiate between short-term market spikes, which typically revert quickly, and prolonged high-stress conditions. The results of the algorithm's maximum likelihood estimation are displayed in Table 1.

The monotonicity of the likelihood function is a key theoretical guarantee of the EM algorithm, ensuring stable convergence behavior during parameter estimation. Dempster et al.'s results in [62] confirm this property for models where the complete data come from the exponential family. Providing this monotonicity to a broader class of Markovian processes significantly expands the applicability of the EM algorithm in time-dependent and state-dependent models.

Theorem 4.2. *The EM algorithm guarantees that the likelihood function for a specified Markovian process does not decrease on each iteration.*

Proof. Let n and m be fixed positive integers, and consider a set $\Xi \subseteq \mathbb{R}^m$. Let X_t be a continuous-time Markov chain with transition probability matrix $P = (p_{ij}) = (p_{e_i e_j})$ and initial distribution vector $p = (p_i) = (p_{e_i})$. Let $\chi(x)$ represent the joint distribution of the vector $(X_{t_0}, \dots, X_{t_n})$ evaluated at the observed points $(x_{t_0}, x_{t_1}, \dots, x_{t_n})$. Let F_t be a stochastic process possessing the Markov property, whose behavior depends on an associated Markov chain X_t . The evolution of F_t is characterized by its conditional transition probabilities given the state of the underlying process. Define

$$\mathbf{P}_{e_i}(t, s; \alpha, \beta) := \mathbb{P}(F_t = \alpha \mid F_s = \beta, X_s = e_i).$$

This measures the probability that F_t takes the value α at time t , given it was β at an earlier time s , assuming the process X_t was in state e_i at time s . Let $\mathcal{G} : \mathbb{R} \times \mathcal{E} \rightarrow \mathbb{R}$ be a function such that, for each $i \in \{1, 2\}$, the corresponding function $g_{e_i}(\alpha, \zeta) := \mathcal{G}(\alpha, e_i, \zeta)$ is a diffeomorphism on $\mathbb{R}$. Let $\tilde{G}_t$ with $\tilde{G}_t = \mathcal{G}(F_t, X_t, \zeta)$. Moreover, define a set of transition probability functions $\varphi := \{\varphi_{e_i e_j}(t, s; u, v)\}_{i,j=1,2}$, where each function is characterized by

$$\varphi_{e_i e_j}(t, s; u, v) := \mathbb{P}(\tilde{G}_t = u \mid \tilde{G}_s = v, X_s = e_i, X_t = e_j)$$

Since $g_{e_j}(u, \zeta)$ is a diffeomorphism, it admits a smooth inverse $g_{e_j}^{-1}(u, \zeta)$. According to $X_t = e_j$ and $X_s = e_i$, we have

$$F_t = g_{e_j}^{-1}(\tilde{G}_t, \zeta), \quad F_s = g_{e_i}^{-1}(\tilde{G}_s, \zeta).$$

In this case, applying Jacobian transformation technique gives

$$\begin{aligned} \varphi_{e_i e_j}(t, s; u, v) &= \mathbb{P}(F_t = g_{e_j}^{-1}(u, \zeta) \mid F_s = g_{e_i}^{-1}(v, \zeta), X_s = e_i, X_t = e_j) \times \left(\frac{d}{du} g_{e_j}^{-1}(u, \zeta) \right) \\ &= \mathbf{P}_{e_i} \left(t, s; g_{e_j}^{-1}(u, \zeta), g_{e_i}^{-1}(v, \zeta) \right) \times \left(\frac{1}{\dot{g}_{e_j}(g_{e_j}^{-1}(u, \zeta), \zeta)} \right), \end{aligned}$$

where the dot in $\dot{g}_{e_j}(u, \zeta)$ denotes the derivative of $g_{e_j}(u, \zeta)$ with respect to u .

Define the probability density function corresponding to the stochastic process $(\tilde{G}_{t_0}, \dots, \tilde{G}_{t_n})$, evaluated at the observed data points $v = (v_0, v_1, \dots, v_n)$, as follows:

$$\begin{aligned} \tilde{P}(P, p, \varphi) \{ \tilde{G}_{t_0} = v_0, \tilde{G}_{t_1} = v_1, \dots, \tilde{G}_{t_n} = v_n \} \\ = \sum_{x_{t_0}, \dots, x_{t_n}} p_{x_{t_0}} p_{x_{t_0} x_{t_1}} \varphi_{x_{t_0} x_{t_1}}(t_1, t_0; v_1, v_0) p_{x_{t_1} x_{t_2}} \varphi_{x_{t_1} x_{t_2}}(t_2, t_1; v_2, v_1) \dots \\ p_{x_{t_{n-1}} x_{t_n}} \varphi_{x_{t_{n-1}} x_{t_n}}(t_n, t_{n-1}; v_n, v_{n-1}). \end{aligned}$$

Assume that for each $\zeta \in \Xi$, the mapping $\zeta \mapsto (P(\zeta), p(\zeta), \varphi(\zeta))$ is continuously differentiable. We represent the marginal over latent states by $\tilde{q}(\zeta)$. In this case, $\tilde{q}(\zeta)$ admits the following representation

$$\tilde{q}(\zeta) = \sum_{x \in \mathcal{E}^{n+1}} \tilde{p}(x, v, \zeta),$$

with

$$\tilde{p}(x, v, \zeta) = p_{x_{t_0}}(\zeta) \prod_{k=1}^n p_{x_{t_{k-1}} x_{t_k}}(\zeta) \varphi_{x_{t_{k-1}} x_{t_k}}(t_k, t_{k-1}; v_k, v_{k-1})$$

$$= p_{x_{t_0}}(\zeta) \prod_{k=1}^n p_{x_{t_{k-1}}|x_{t_k}}(\zeta) \times \frac{\mathbf{P}_{x_{t_{k-1}}} \left(t_k, t_{k-1}; g_{x_{t_k}}^{-1}(v_k, \zeta), g_{x_{t_{k-1}}}^{-1}(v_{k-1}, \zeta) \right)}{\dot{g}_{x_{t_k}}(g_{x_{t_k}}^{-1}(v_k, \zeta), \zeta)}.$$

For $(\zeta, \zeta') \in \Xi \times \Xi$, we define the following function

$$L(\zeta, \zeta') := \mathbb{E}_x [\log(\tilde{p}(x, v, \zeta)) | \tilde{G} = v, \zeta'] = \sum_x \mu_v(x, \zeta') \log(\tilde{p}(x, v, \zeta)),$$

such that

$$\mu_v(x, \zeta') := \frac{\tilde{p}(x, v, \zeta')}{\tilde{q}(\zeta')},$$

is the conditional distribution of X given the observed data $\tilde{G}$. Let $\tilde{p}_x(v, \zeta)$ denotes the conditional probability distribution of the observation sequence $\tilde{G}$, given the state process X and parameter ζ . As a result, we obtain the following representation

$$\begin{aligned} \Delta(\zeta, \zeta') &:= \log(\tilde{q}(\zeta)) - \log(\tilde{q}(\zeta')) \\ &= \log \left(\sum_x \tilde{p}_x(v, \zeta) \chi(x, \zeta) \right) - \sum_x \mu_v(x, \zeta') \log(\tilde{q}(\zeta')) \\ &= \log \left(\sum_x \frac{\tilde{p}_x(v, \zeta) \chi(x, \zeta)}{\mu_v(x, \zeta')} \mu_v(x, \zeta') \right) - \sum_x \mu_v(x, \zeta') \log(\tilde{q}(\zeta')) \end{aligned}$$

Because $\mu_v(x, \zeta')$ defines a probability distribution over x with the property $\sum_x \mu_v(x, \zeta') = 1$, we can apply Jensen's inequality to derive the subsequent inequality

$$\begin{aligned} \Delta(\zeta, \zeta') &\geq \sum_x \mu_v(x, \zeta') \log \left(\frac{\tilde{p}_x(v, \zeta) \chi(x, \zeta)}{\mu_v(x, \zeta')} \right) - \sum_x \mu_v(x, \zeta') \log(\tilde{q}(\zeta')) \\ &= \sum_x \mu_v(x, \zeta') \log \left(\frac{\tilde{p}_x(v, \zeta) \chi(x, \zeta)}{\mu_v(x, \zeta') \tilde{q}(\zeta')} \right) \\ &= \sum_x \mu_v(x, \zeta') \log \left(\frac{\tilde{p}(x, v, \zeta)}{\tilde{p}(x, f, \zeta')} \right) =: \delta(\zeta, \zeta'). \end{aligned}$$

Suppose

$$\zeta'' := \arg \max_{\zeta} \delta(\zeta, \zeta') = \arg \max_{\zeta} L(\zeta, \zeta').$$

In this case,

$$\log(\tilde{q}(\zeta'')) - \log(\tilde{q}(\zeta')) = \Delta(\zeta'', \zeta') \geq \delta(\zeta'', \zeta') \geq \delta(\zeta', \zeta') \geq 0.$$

The previously derived expression guarantees that the likelihood function is non-decreasing at each step of the EM algorithm. Such monotonicity of the likelihood function at every iteration is a fundamental characteristic of the EM procedure, which ensures convergence to a local optimum.

To present the results of parameter estimation, we set the stopping criterion of the EM algorithm such that the mean squared error (MSE) between the current and previous parameter estimates is less than 10^{-3} . The estimated parameters obtained at different iterations of the EM algorithm are shown in Fig. 1. Moreover, this figure illustrates the convergence of the parameter estimation procedure.

Fig. 2 provides an empirical illustration of the proposed regime switching model. The left panel shows the evolution of United States gas futures prices from January 1, 2017, to January 1, 2023, revealing multiple episodes of sharp fluctuations and relative calm. The right panel presents the regime classification derived from the model, where each trading day is assigned to either the low-volatility regime or the high-volatility regime. Notably, prolonged periods of stability are correctly identified as e_2 , while episodes of significant market stress — such as those during geopolitical tensions or extreme weather events — are captured by e_1 . This classification confirms the model's ability to differentiate between transient price shocks and persistent volatility, offering valuable insights for energy market participants and risk managers.

4.1. Forecasting performances

We now assess the forecasting capabilities of the proposed Time-Varying Volatility Model with regime switching (TVRS) and compare it against the regime switching Black-76 (RSB-76) model and the Moving Average with lag 10 (MA(10)) model. The use of a moving average with lag 10 is primarily based on empirical tuning. During preliminary experiments, we tested various lag values (e.g., 5, 10, 15, 20) and found that a lag of 10 provided a good balance between smoothing short-term fluctuations and preserving relevant signal dynamics in the data. While this choice is not directly derived from theoretical principles, it aligns with common practice in time series analysis and financial modeling in this study, where moderate lag lengths (such as 10) are employed to capture short-term trends without introducing excessive delay.

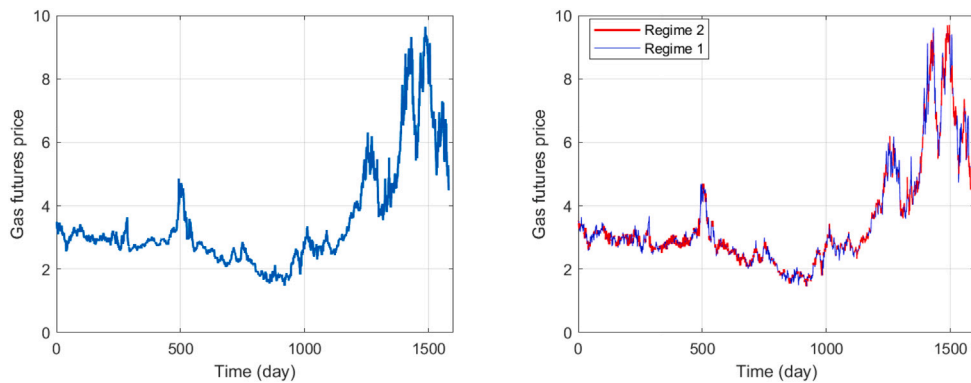


Fig. 2. Left: Gas futures prices in the United States from January 1, 2017, to January 1, 2023. Right: Classification of these prices according to the proposed regime switching energy model.

For the TVRS model, the one-step ahead forecast of F_t is given by:

$$\begin{aligned} \mathbb{E}[F_{t+1}|F_t] = & F_t p_{11} \mathbb{P}(X_t = e_1|F_t) + F_t p_{22} \mathbb{P}(X_t = e_2|F_t) \\ & + F_t p_{21} \mathbb{P}(X_t = e_1|F_t) + F_t p_{12} \mathbb{P}(X_t = e_2|F_t). \end{aligned} \quad (21)$$

Furthermore, the d -step ahead predicted probabilities are obtained using:

$$\mathbb{P}(X_{t+d} = e_j|F_t) = \mathcal{P}^d \mathbb{P}(X_t = e_j|F_t),$$

where $\mathcal{P} = (p_{ji})_{i,j=1,2}$ denotes the regime's transition matrix, that is $p_{ij} = \mathbb{P}(X_t = e_j|X_{t-1} = e_i)$, and $\mathcal{P}^d$ is the d th power of $\mathcal{P}$. In particular, for $d = 1$, we have:

$$\mathbb{P}(X_{t+1} = e_1|F_t) = p_{11} \mathbb{P}(X_t = e_1|F_t) + p_{12} \mathbb{P}(X_t = e_2|F_t),$$

and

$$\mathbb{P}(X_{t+1} = e_2|F_t) = p_{21} \mathbb{P}(X_t = e_1|F_t) + p_{22} \mathbb{P}(X_t = e_2|F_t).$$

We conducted a one-step ahead forecast for the above-mentioned models using gas futures prices in the United States from January 1, 2017, to January 1, 2023, and carried out recursive forecasts from January 1, 2023, to March 1, 2023. The most commonly used forecast error measures, namely the MSE, the Root Mean Square Error (RMSE), and the Mean Absolute Error (MAE), were used to assess model performance. The results, presented in Table 2, indicate that our proposed TVRS model consistently outperforms other models across all forecast horizons considered in terms of MSE, RMSE, and MAE. Further comparisons of actual versus predicted gas futures prices by the proposed TVRS model, starting from January 1, 2023, are illustrated in Fig. 3. The MSE values recorded for the TVRS model over four forecast horizons (5, 15, 25, and 35 days) are 0.0152, 0.0277, 0.0187, and 0.0319, respectively. In contrast, the MSE values for the RSB-76 model and the MA(10) model for these same horizons are significantly higher, reported as [1.2348, 7.0387, 7.4694, 7.1855] for the RSB-76 model, and [1.0414, 1.6391, 2.8040, 3.7087] for the MA(10) model, respectively.

We performed the Diebold–Mariano test, as described in [68], to analyze the statistical significance of variations in MSE and MAE between forecasting models. The null hypothesis of the Diebold–Mariano test states that the two models under comparison provide forecasts of equivalent accuracy, suggesting that any observed differences in their forecast errors are statistically insignificant. Conversely, the alternative hypothesis asserts that one model outperforms the other in terms of forecasting accuracy. By applying the Diebold–Mariano test to both MSE and MAE measures, we aim to demonstrate that our proposed TVRS model yields more accurate forecasts than both the MA(10) model and the RSB-76 model.

The results of the Diebold–Mariano test are detailed in Table 3. The null hypotheses, asserting equivalent forecasting accuracy between the models, cannot be rejected at a significance level of 5%. However, at a significance level of 4%, the null hypothesis is rejected for the short time forecast horizons, indicating superior forecasting accuracy of the TVRS model over the RSB-76 model. Similarly, at a significance level of 6%, the TVRS model demonstrates better forecasting performance than the MA(10) model for the same forecast horizon. For longer time forecast horizons, neither null hypothesis is rejected at a significance level of 0%, suggesting approximate equivalence in forecasting accuracy. Despite this, the overall analysis confirms that the proposed TVRS model generally surpasses the MA(10) and RSB-76 models across various forecast horizons. The incorporation of time-varying volatility enhances the TVRS model's capability to effectively capture the dynamics of energy futures, such as gas futures, making it particularly valuable for extended forecasting periods. This improved forecasting accuracy over longer durations is essential for robust energy risk management, a critical component for traders in the energy sector. Given these findings, we recommend that energy traders consider adopting the TVRS model to enhance their risk management strategies.

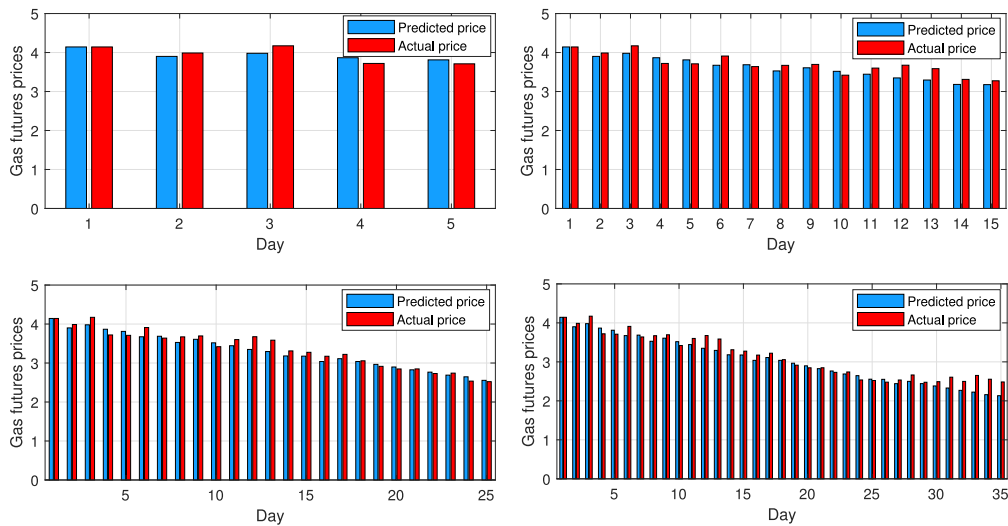


Fig. 3. Comparing the actual gas futures prices in United States starting from 01 Jan 2023 with the predicted prices by 1-step ahead method implemented by the TVRS model.

Table 2
Forecast performance of the models with various time horizons.

Days	TVRS model			RSB-76 model			MA(10) model		
	MSE	RMSE	MAE	MSE	RMSE	MAE	MSE	RMSE	MAE
5	0.0152	0.1234	0.1054	1.2348	1.1112	0.8253	1.0414	1.0205	1.0027
10	0.0172	0.1312	0.1140	5.0059	2.2374	1.8900	1.3460	1.1602	1.1371
15	0.0277	0.1663	0.1428	7.0387	2.6531	2.3700	1.6391	1.2803	1.2552
20	0.0225	0.1501	0.1254	7.5522	2.7481	2.5306	2.1711	1.4735	1.4274
25	0.0187	0.1369	0.1105	7.4694	2.7330	2.5583	2.8040	1.6745	1.6048
30	0.0174	0.1320	0.1079	7.2907	2.7001	2.5533	3.3427	1.8283	1.7478
35	0.0319	0.1786	0.1405	7.1855	2.6806	2.5541	3.7087	1.9258	1.8459
40	0.0364	0.1907	0.1545	6.9300	2.6325	2.5182	4.1729	2.0428	1.9563

Table 3
Diebold–Mariano test.

Days	TVRS model versus RSB-76 model		TVRS model versus MA(10) model	
	Test statistic	p-value	Test statistic	p-value
5	1.7272	4.2063e–02	2.4781	6.6049e–03
10	3.7441	9.0506e–05	6.1356	4.2412e–10
15	6.0127	9.1234e–10	9.0928	4.8243e–20
20	8.4079	2.0879e–17	10.981	2.3513e–28
25	10.406	1.1599e–25	11.673	8.7803e–32
30	12.128	3.7719e–34	12.289	5.1878e–35
35	13.873	4.5869e–44	13.203	4.2264e–40
40	14.858	3.0743e–50	13.372	4.413e–41

5. Numerical results

Here we examine the sensitivity of our European option pricing formula to changes in the parameters of the TVRS model. But first, in addition to the mathematical derivation of our option price formula in [Theorem 3.1](#), we further validate its accuracy numerically through generalized antithetic variates Monte-Carlo simulations, a robust benchmarking method recently proposed by [\[43\]](#). This technique, initially developed for pricing European options, enhances the traditional antithetic variates method by increasing the number of variates from two units to at least three units. In [\[43\]](#), the authors developed the generalized antithetic variates estimator (GAVE), and demonstrated that as the number d of antithetic variates increases, the variance of the GAVE significantly diminishes, directly resulting in substantial improvements in the performance of Monte-Carlo simulations (for details on the GAVE structure, refer to [Appendix B](#)).

We now present a sensitivity analysis of the spot volatility, intensity, mean reversion speed, and weight parameters on the European option price of energy futures under the TVRS model [\(7\)](#). For this analysis, we set the initial energy futures, strike prices

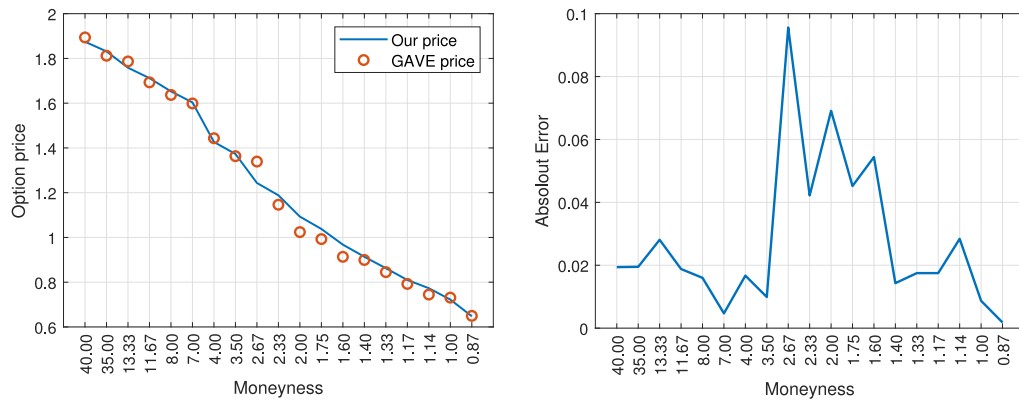


Fig. 4. Comparing the European option price written on energy futures obtained from Theorem 3.1 and GAVE method.

and interest rate at $f = 3.5$ and $K = 3$ and $\bar{r} = (0.02, 0.04)^\top$, respectively. The remaining parameters are drawn from those estimated based on the gas futures prices in the United States from January 1, 2017, to January 1, 2023, as reported in Table 1. Additionally, we assume the weight parameter α to be 0.5, indicating that the option price is the average of prices in regimes 1 and 2, expressed as $U = 0.5U_1 + 0.5U_2$.

We also numerically corroborate the accuracy of our semi-closed form formula. In Fig. 4, we present a comparison between the option prices derived from our formula and those obtained using the GAVE simulation method. The analysis is based on $d = 16$ antithetic variates, $m = 1000$ Monte-Carlo simulations, and $n = 100$ discretizations. Fig. 4 (left) shows the comparison in terms of moneyness, that is the value of the energy futures option prices relative to the strike price. The comparison reveals a close point-wise alignment between the two sets of prices across all moneyness values. Furthermore, as illustrated in Fig. 4 (right), the absolute error between our pricing and the GAVE pricing remains below 10%, demonstrating the accuracy and practical reliability of our semi-closed form formula. The absolute error criterion used in this study is defined as follows

$$\varepsilon = |\text{Our price} - \text{GAVE price}|.$$

We compare the computational efficiency of our proposed option pricing model for energy futures, derived through a semi-analytical formula as established in Theorem 3.1, with the GAVE method. The comparison, illustrated in Fig. 5, highlights a significant reduction in computational time when using the semi-analytical option pricing model in Theorem 3.1. This improvement is attributed to the analytical nature of the proposed approach, which avoids the intensive sampling and simulation steps required by the GAVE method.

We additionally examine how regime switching parameters influences the pricing of options. Assuming equal transition rates for both regimes, we have depicted option prices in relation to spot volatility for regimes 1 and 2 in Fig. 6. Consistent with expectations, the TVRS model demonstrates that option prices increase with rising spot volatility. This increase reflects the higher potential range of outcomes for the underlying security, which leads to higher option prices, indicating significant potential fluctuations in the security's price over short periods. Furthermore, as detailed in Table 1, spot volatility is higher in regime 1 than in regime 2, reflecting that the dynamics of the futures contract are more volatile in economically unfavorable conditions. Consequently, it becomes evident that buyers should expect to pay more for a European call option on energy futures when the economy is in a poorer state. This relationship is also clearly apparent in Fig. 6.

Fig. 7 presents option prices under varying intensity parameters between regimes 1 and 2. As the intensity parameter λ_1 increases, enhancing the transition likelihood from state e_1 to e_2 , the price of options in regime 1 decreases. This reflects the previous observation that European options are costlier in state e_1 and less expensive in state e_2 . Conversely, an increase in λ_2 , which heightens the transition likelihood from state e_2 to state e_1 , leads to higher option prices in regime 2. This is due to the higher costs associated with European options in state e_1 compared to state e_2 .

According to mean reversion theory from financial literature, both asset price volatility and historical returns tend to revert to the long-term mean or average level of the complete dataset over time, with mean reversion speed dictating the pace of this return. It is clear from Eq. (8) that an increase in mean reversion speed results in a decreased market volatility. Consequently, it can be inferred that the option price decreases as the mean reversion speed increases, a relationship that is evident in the left subplot of Fig. 8. In the right subplot of Fig. 8, the impact of the weight parameter α is analyzed. As previously discussed, the option price is higher in regime 1 than in regime 2, that is $U_1 \geq U_2$. As the option price U is the weighted average of U_1 and U_2 , given by $U = \alpha U_1 + (1 - \alpha)U_2$, an increase in α results in an increase in the option price U .

From Fig. 9, as the strike price increases, the futures position γ in the hedging portfolio decreases at time $t = 0$. In the case of a call option, as the strike price increases, the likelihood that the option will expire in the money decreases, reducing its delta. Since the hedge using futures is constructed to offset the option's delta, a lower delta necessitates a smaller futures position. This relationship reflects the diminishing influence of the underlying asset on the option's value as the strike becomes less attainable, thereby reducing the amount of futures needed to maintain an effective hedge. Moreover, according to Fig. 9, with increasing strike

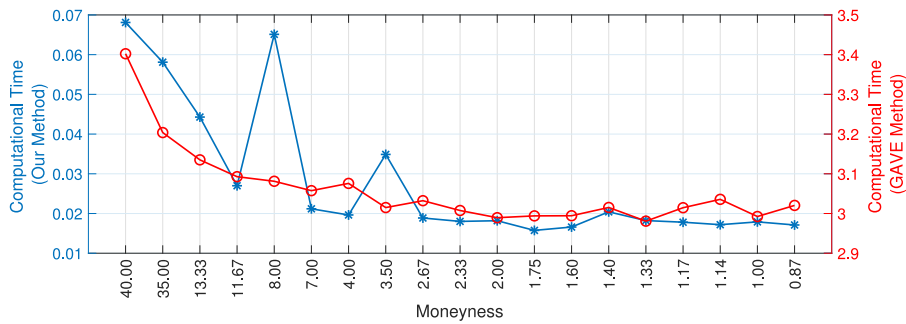


Fig. 5. Comparing the computational time (seconds) for pricing European options on energy futures using Theorem 3.1 and the GAVE method.

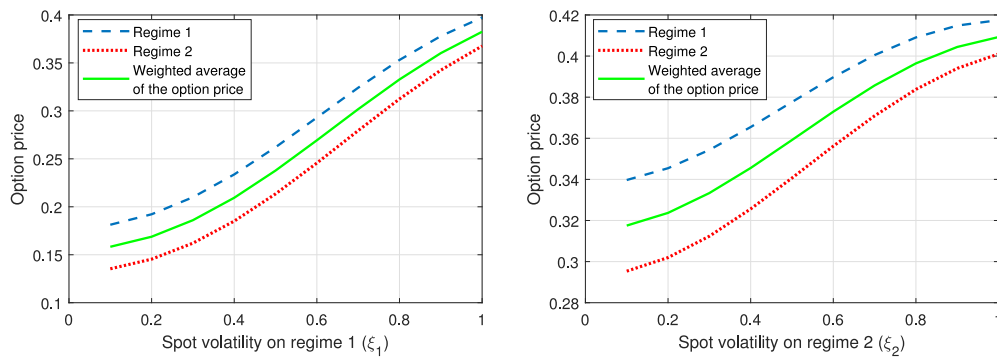


Fig. 6. European option price written on energy futures with changing spot volatility on regimes 1 and 2.

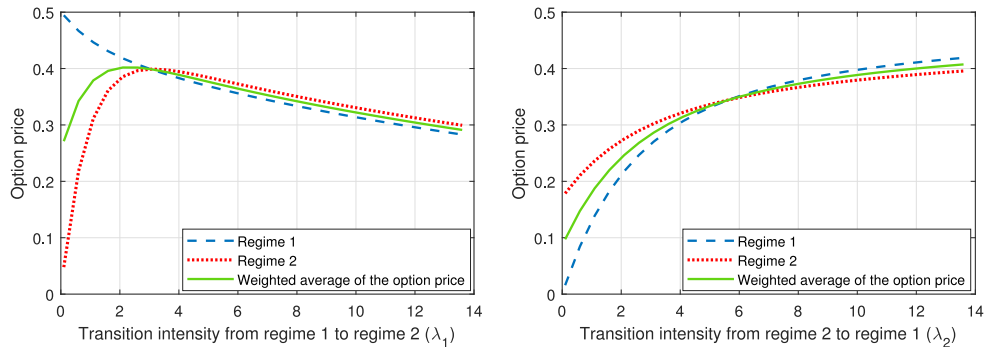


Fig. 7. European option price written on energy futures with changing transition intensity parameters λ_1 and λ_2 .

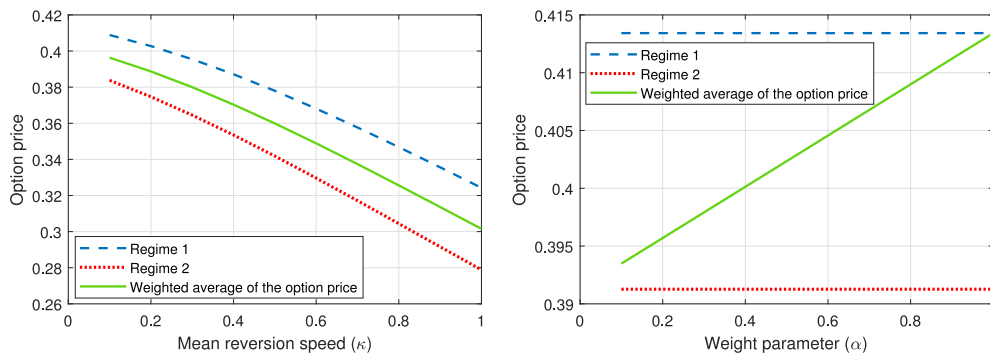


Fig. 8. European option price written on energy futures with changing mean reversion speed parameter κ and weight parameter α .

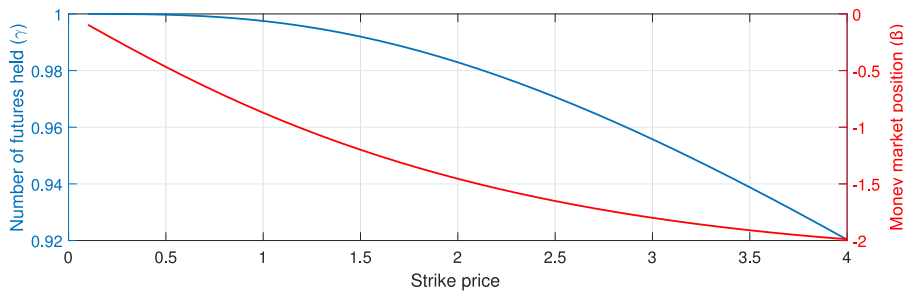


Fig. 9. The impact of strike price on the two components of the hedging strategy.

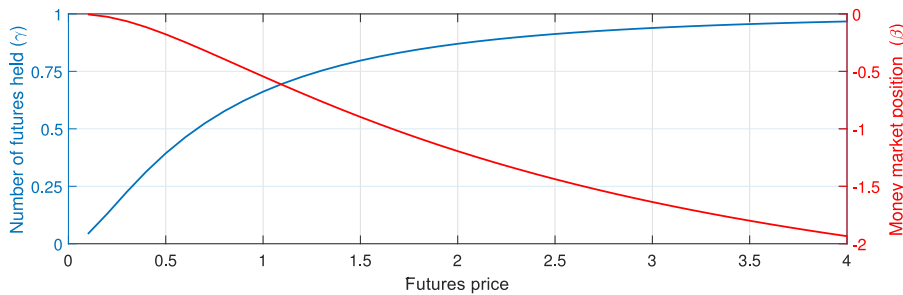


Fig. 10. The impact of futures price on the two components of the hedging strategy.

price, the money market position β decreases at time $t = 0$. This reflects the need for increased borrowing to fund the hedge as the strike price rises.

As the initial futures price surpasses the strike price by a significant margin (deep in-the-money), the option behaves almost like owning the underlying asset itself, and its delta approaches a maximum value (see Fig. 10). At this stage, further increases in the initial futures price cause only very small changes in γ . Meanwhile, from this figure, β decreases since a more valuable option requires additional borrowing to finance the larger futures position needed for the hedge.

6. Conclusion remarks

In this paper, we have investigated energy futures using a time-varying volatility model enhanced by a regime switching factor. The proposed model captures structural shifts in market dynamics from stressed to non-stressed states. After finding the appropriate pricing kernel using a generalized version of Esscher transforms, we derived a semi-closed form formula for the price of European call options and suggested a mean-self-financing hedging strategy for these options. Generalized antithetic variates Monte-Carlo simulations were used to numerically validate our formula. We have studied the impact of regime switching risk factors on option prices. Numerical examples are provided to illustrate the significance of spot volatility, state transition intensity, mean reversion speed, and weight parameters on these prices. To carry out an empirical study, we calibrated our model from real gas futures price data using the EM algorithm. Our empirical findings demonstrate that the time-varying volatility model with a regime switching factor surpasses the regime switching Black-76 model and the moving average model in explaining and forecasting gas futures price movements.

CRedit authorship contribution statement

Guillaume Leduc: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Farshid Mehrdoust:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Idin Noorani:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Acknowledgments

Guillaume Leduc was supported by a Faculty Research Grant (FRG24-E-S17) from the American University of Sharjah.

Appendix A. Laplace cumulant process

We recall some definitions from [44]. Consider a *special* semi-martingale Y_t , meaning a semi-martingale that admits a unique decomposition (up to indistinguishability) into the sum of a local martingale and a finite variation process, and suppose $Y_0 = 0$. Then there exists (up to indistinguishability) a semi-martingale L_t such that

$$\exp(Y_t) = 1 + \int_0^t \exp(Y_{s-}) dL_s. \quad (22)$$

The semi-martingale L_t is called the *logarithmic transform* of Y_t . Conversely Y_t is called the *exponential transform* of L_t , and it is the unique (up to indistinguishability) process such that Eq. (22) is satisfied. Then the predictable part of finite variation, M_t , of L_t is called the *Laplace cumulant process* of Y_t .

Appendix B. Generalized antithetic variates estimator

Let n , m , and d be positive integers. Define Γ_n^{md} as the union of the sets $\{y_1^{jk}, \dots, y_n^{jk}\}$ for $j = 1$ to m and $k = 1$ to d . Here, n denotes the number of discretization steps, m the number of naive Monte-Carlo simulations, and d the number of antithetic variates. Γ_n^{md} represents standard normal variables in $\mathbb{R}^{n \times m \times d}$, with $\text{Cov}[y_i^{jk}, y_{i'}^{j'k}] = 0$ for all $i \neq i'$, $\text{Cov}[y_i^{jk}, y_i^{j'k}] = 0$ for all $j \neq j'$, and $\text{Cov}[y_i^{jk}, y_i^{jk'}] < 0$ for all $k \neq k'$, where $i, i' \in \{1, \dots, n\}$, $j, j' \in \{1, \dots, m\}$, and $k, k' \in \{1, \dots, d\}$. This structure, proposed by [43], facilitates the representation of asset trajectories, specifically energy futures, using n discretizations, m Monte-Carlo simulations, and d antithetic normal random vectors. It follows the following scheme:

$$\begin{cases} \mathbf{F}_{t_1}^j = (F_{t_1}^{j1}, F_{t_1}^{j2}, \dots, F_{t_1}^{jd}), \text{ with } \text{Cov}[F_{t_1}^{jk}, F_{t_1}^{jk'}] < 0 \text{ for } k, k' = 1, \dots, d \text{ and } k \neq k' \\ \vdots \\ \mathbf{F}_{t_n}^j = (F_{t_n}^{j1}, F_{t_n}^{j2}, \dots, F_{t_n}^{jd}), \text{ with } \text{Cov}[F_{t_n}^{jk}, F_{t_n}^{jk'}] < 0 \text{ for } k, k' = 1, \dots, d \text{ and } k \neq k', \end{cases}$$

where for any $i = 1, \dots, n$, $j = 1, \dots, m$, $k = 1, \dots, d$,

$$\begin{aligned} F_{t_i}^{jk} &= F_{t_{i-1}}^{jk} + \sigma^j(t_i, T, X_{t_i}) F_{t_{i-1}}^{jk} \sqrt{\delta} y_i^{jk}, \\ \sigma^j(t_i, T, X_{t_i}) &= \xi_i^j \sqrt{\frac{1 - e^{-2\kappa(T-t_i)}}{2\kappa(T-t_i)}}, \end{aligned}$$

and,

$$\text{Cov}[F_{t_i}^{jk}, F_{t_i}^{j'k}] = 0, \text{ and } \text{Cov}[F_{t_i}^{jk}, F_{t_i'}^{jk}] = 0,$$

for $i, i' = 1, \dots, n$, $j, j' = 1, \dots, m$, and $k = 1, \dots, d$. Defining

$$\mathbf{F} = \bigcup_{j=1}^m \bigcup_{k=1}^d \bigcup_{i=0}^n F_{t_i}^{jk},$$

results in $m \times d$ simulated paths, with $m \times \binom{d}{2}$ of them being negatively correlated. As proven in [43], $\Theta^d := \frac{e^{-rT}}{md} \sum_{j=1}^m \sum_{k=1}^d \max(F_{t_n}^{jk} - K, 0)$ is an unbiased estimator for $e^{-rT} \mathbb{E}[\max(F_T - K, 0)]$, with $\text{Var}[\Theta^{d+1}] < \text{Var}[\Theta^d]$. The following algorithms determine the status of the parameters of the model and the time of regime state changes.

Algorithm 1: Simulation of the parameters of model and time of the state changes.

Step 1. Define the initial state X_0 at state q , where $q \in \{e_1, e_2\}$.

Step 2. Simulate the vector of state changes $0 = \tau_0 < \tau_1 < \dots < \tau_{H+1} = T$ by drawing independent exponential random variables $\mathcal{T}_i := \tau_{i+1} - \tau_i$, for all $i = 1, \dots, H$, where $\mathcal{T}_i \sim \text{Exp}(-\mathbb{P}(X_{\tau_i}, X_{\tau_i}))$. This is repeated until $\min(\tau_i, T) = T$.

Step 3. For $i = 1, \dots, n$ if we are in the state 1 at time t_i , we set $(r_i, \xi_i) = (r_1, \xi_1)$; else we set $(r_i, \xi_i) = (r_2, \xi_2)$.

Returning to discretization version of model (17), Algorithm 3 provides the M simulation of futures path and numerical procedure for valuing European option under the commodity futures model (17). Using the properties of a Markov chain, this algorithm simulates the time to the next state change $\mathcal{T}_i := \tau_{i+1} - \tau_i$ by a exponential random variable, i.e. $(\mathcal{T}_i | X_{\tau_i} = e_s) \sim \text{Exp}(-\Pi(s, s))$, for $s = 1 \vee 2$. Algorithm 2 illustrates the m step simulated and n discretized point. Suppose that H is the number of jump. Since the parameters of model associated with the regime switching and change over time, hence before propose the algorithm of European option pricing, we present the simulation of the parameters of model and time of the state changes τ_i , for all $i = 1, \dots, H$ in Algorithm 1. The following Algorithm provides the d standard normal random variable with negative correlation by Cholesky method:

Algorithm 2: Generate standard normal random variable with negative correlation.

Step 1. Repeat Step (2)-(5) until $L_{d \times d}$ is positive definite.

Step 2. For $j = 1, \dots, d$

Step 3. Set $L_{j,j} = 1$

Step 4. For $i = j + 1, \dots, d$
 Step 5. Set $L_{i,j} = -a$ and $L_{j,i} = L_{i,j}$ (w.l.o.g. let a be an independent uniform random variable on $[0, 1]$)
 Step 6. For $j = 1, \dots, d$
 Step 7. For $i = j, \dots, d$
 Step 8. Set $u_i = L_{i,j}$
 Step 9. For $k = 1, \dots, j - 1$
 Step 10. Set $u_i = u_i - L_{j,k} L_{i,k}$, and $\mathcal{L}_{i,j} = \frac{u_i}{\sqrt{u_j}}$
 Step 11. Draw the vector of independent standard normal random variables $(y_1, \dots, y_d)$
 Step 12. For $i = 1, \dots, d$
 Step 13. For $j = 1, \dots, d$
 Step 14. Set $Y_i = \mathcal{L}_{i,j} y_j$

For fixed n and m , let Γ_n^{md} be the sequence of samples defined by $\Gamma_n^{m(d+1)} = \Gamma_n^{md} \cup \{y_i^{j(K+1)}\}$, for all $i = 1, \dots, n$, and $j = 1, \dots, m$. We consider the Θ^d defined on the sample Γ_n^{md} as follows

$$\Theta^d = \frac{1}{dm} \sum_{j=1}^m \sum_{k=1}^d r^{jk} \max(F_{t_n}^{jk} - K, 0). \quad (23)$$

It is clear that the problem (23) for the case $d = 1$ correspond to the naive Monte-Carlo simulation and for $d = 2$ is similar to the standard antithetic variates. Algorithm 4 provide the $d \times m$ simulation of futures path and numerical procedure for valuing European option by GAVE method under the propose futures model (17).

Algorithm 3: European option pricing under the commodity futures model (17). This algorithm estimates the value of the European option, by the GAVE method.

Step 1. Repeat Step (2)-(6) until $j \leq m$, ($j = 1, 2, \dots, m - 1$). Then proceed the Step (7).
 Step 2. Repeat Step (3)-(5) for $k = 2, \dots, d$. Then proceed with Step (6).
 Step 3. Simulation of parameters $(r_{t_i}^j, \xi_{t_i}^j)_{i=1, \dots, n}$ and vector of state changes by Algorithm 1.
 Step 4. Draw a vector of independent standard normal variables $(y_1^j, \dots, y_n^j)$, independent of $\mathcal{T}_t$, for $i = 1, \dots, H$.
 Step 5. Draw a vector of standard normal variables $(y_1^{jk}, \dots, y_n^{jk})$. Such that $\text{cov}(y_i^{jk}, y_{i'}^{jk}) = 0$, $\text{cov}(y_i^{jk}, y_{i'}^{j'k}) = 0$ and $\text{cov}(y_i^{jk}, y_{i'}^{jk'}) < 0$ for any $i \neq i'$, $j \neq j'$, and $k \neq k'$ where $i, i' \in \{1, \dots, n\}$, $j, j' \in \{1, \dots, m\}$ and $k, k' \in \{1, \dots, d\}$. This Step is performed by Algorithm 2.
 Step 6. For $i = 1, \dots, n$, simulate the commodity futures value $F_{t_i}^{jk}$ and discounted factor $e^{-\int_0^t r_v dv}$, that indexed by the current simulation $j = 1, 2, \dots, m$, and $k = 1, 2, \dots, d$, where

$$F_{t_i}^{jk} = F_{t_{i-1}}^{jk} + \sigma^j(t_i, T, X_{t_i}) F_{t_{i-1}}^{jk} \sqrt{\delta} y_i^{jk},$$

$$r^{jk} = \exp\left(-\sum_{i=0}^H (\tau_{i+1} - \tau_i) \times r_{t_i}^j\right).$$

Step 7. Estimate the option price at initial state q by

$$\Theta^{q-d} := \frac{1}{dm} \sum_{j=1}^m \sum_{k=1}^d r^{jk} \max(F_{t_i}^{jk} - K, 0).$$

Note. In accordance with Step 1 of Algorithm 1, and by re-executing Algorithms 1-3, the estimation of European option price using GAVE method is expressed as follows

$$\tilde{\alpha} \Theta^{q-d} \Big|_{q=e_1} + (1 - \tilde{\alpha}) \Theta^{q-d} \Big|_{q=e_2}.$$

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